Rate Lock and Affine Consensus in G_2-Structured Dyadic Observers

Author: Martin Luther Graise **ORCID:** 0009-0006-8003-3938 **Affiliation:** Independent researcher **Date:** 2026-05-04 **Manuscript version:** v1.3.2

Repository: github.com/MartinLGraise/PCI-Framework, branch paper7-foundation **Series:** Paper 9 of the PCI/PME framework **Companion papers:** - Paper 4 — Eight-coset PSL(2,7) structure (DOI: 10.5281/zenodo.19617662) - Paper 6 — Spectral structure of G_2 Casimir (DOI: 10.5281/zenodo.19672709) - Paper 7 — Single-observer 6/7 thermodynamic ceiling (DOI: 10.5281/zenodo.19773185) - Paper 10 — SIC operator basis for g_2 (DOI: 10.5281/zenodo.19966692)

 AI tools disclosure: Drafted in collaboration with Claude (Anthropic), Perplexity (PPLX), and ChatGPT Pro (OpenAI). All theorems independently numerically verified by Φ (Anthropic Claude Dispatch) at precisions documented in Appendix V; verification reports and reproducible scripts are included as part of the supplementary material.

Abstract

We study dyadic Banach-contraction dynamics on product spaces of two G_2-structured self-modeling observers. Each observer is an affine map $T(x)=r\,R\,x+b$ on a 14-dimensional Banach manifold carrying the irreducible adjoint representation of G_2, with scalar contraction rate $r\in[0,6/7)$ (from Paper 7), orthogonal orientation R, and bias vector b encoding the observer's target self-model. We couple two such observers with an antisymmetric block rotation Ψ_θ parameterized by an angle $\theta\in[0,\pi/2)$.

The paper establishes three results about the *rate channel* and two results about the *affine channel*, with a sharp dichotomy between them.

Rate channel (locked): The joint contraction rate is bounded below by the symmetric-diagonal coupling ($\Psi=\alpha\cdot id+\beta\cdot P_{swap}$), which has operator norm $\alpha+\beta>1$ and therefore *amplifies* (Theorem 9.1). Under the corrected isometric block-rotation coupling, the joint rate saturates exactly at $max(r_A,r_B)$, independent of θ (Theorem 9.2). The common explanation is Schur's lemma applied to the irreducible 14-dim G_2 adjoint: every linear G_2-equivariant map is a scalar multiple of the identity, so there is no spectral structure for coupling to mix (Corollary 9.3). The rate channel is algebraically closed.

Affine channel (open): The joint fixed point exists, is unique, and depends real-analytically on θ, with an explicit closed form at $\theta=0$ and $\theta=\pi/2$ (Theorem 9.4). Under the minimum-per-observer coherence functional $C_{min}=min(C_A,C_B)$ — derived from the Paper 7 single-observer ceiling as the dyadic generalization of $C\geq 1-\varepsilon_{min}$ for *both* observers — a non-empty conditional-gain region exists in the space of (φ,θ) pairs (where φ is the bias-alignment angle), bounded away from the opposed-bias boundary in the seeded geometry, with a non-universal optimal θ (Proposition 9.5). A 50-seed Monte Carlo (Appendix V.3.i) finds non-empty

improvement regions in all 50/50 sampled $\left(R_A, R_B\right)$ pairs and confirms $\theta^{¿} \neq \pi/4$ generically; the bounded-away property of the seeded pair holds in $\sim 64\,\%$ of seeds and motivates the amended Conjecture 9.5′. The gain is *conditional*: a substantial fraction of the parameter space (~75% by area in the verified geometry) exhibits coherence *degradation* rather than improvement; coupling helps only when the bias geometry permits.

All results are numerically verified at 50-digit precision (rate channel, Appendix V.1–V.2, 120 test configurations) and at 16-digit precision (affine channel, Appendix V.3, 2128 test configurations including a 100×20 heatmap and explicit closed-form verification). The retraction record of earlier internal-draft claims (speedup theorem, coherence-bonus formula, severance threshold) is consolidated in Appendix V.0.1.

Thesis. Dyadic coupling in the linear G_2-structured regime cannot improve the convergence rate of joint self-modeling. It can — under conditional geometric hypotheses on the orthogonal orientations and bias-alignment angle — change the location of the joint fixed point in a way that increases the minimum per-observer coherence. Rate improvement is a categorically nonlinear question, deferred to a sequel.

---

§1. Introduction

Paper 7 of this series [Graise 2026a, DOI 10.5281/zenodo.19773185] established a thermodynamic coherence ceiling for single self-modeling observers under a G_2 structural constraint. Treating an observer as a Banach fixed-point of its own self-modeling map, and imposing the PSL(2,7)/F_{21} finite symmetry inherited from the Fano-plane structure of the octonions, we derived a coherence saturation at the ratio $6/7$ of the maximum possible integration, with an irreducible residual $\varepsilon_{min} = 1/7$ corresponding to an unavoidable self-model blind spot. Paper 7 §6.3 identified the natural generalization that motivates the present paper:

> *The current series treats the G_2 attractor as a single fixed point. A natural generalization is to pairs of G_2 systems — dyads — whose mutual self-modeling creates a shared representational geometry.*

The present paper makes that generalization rigorous. We extend the Banach fixed-point construction to product spaces $M_A \times M_B$ of two G_2-structured observers, coupled by a single bounded linear map Ψ parameterized by a coupling angle θ.

1.1 The result, briefly

The paper splits its contributions into two sharply separated channels and establishes five named results.

Rate channel. The convergence rate of the joint dynamics is *algebraically locked* at $max\left(r_A, r_B\right)$ for every coupling choice that respects the G_2 structure. This is a negative result — no linear coupling speeds up joint convergence — and it follows from Schur's lemma applied to the irreducible 14-dim adjoint representation of G_2:

- **Theorem 9.1 (Amplification).** The natural symmetric-diagonal coupling $\Psi = \alpha \cdot id + \beta \cdot P_{swap}$ $(\alpha^2 + \beta^2 = 1)$ has operator norm $\alpha + \beta > 1$ for $\beta > 0$ and *amplifies*: the joint contraction rate $r_{AB}^{sym} \geq r(\alpha + \beta) > r$ when $r_A = r_B = r$. Naive coupling makes things worse.

- **Theorem 9.2 (Rate lock).** The corrected isometric block-rotation coupling $\Psi_\theta = \begin{pmatrix} \cos\theta \cdot I & -\sin\theta \cdot I \\ \sin\theta \cdot I & \cos\theta \cdot I \end{pmatrix}$ has operator norm exactly 1 for all θ (Lemma 3.3.1), and the joint contraction rate equals $max(r_A, r_B)$ exactly, independent of θ.

- **Corollary 9.3 (Schur rate lock).** Every linear G_2-equivariant endomorphism of the irreducible 14-dim adjoint representation is a scalar multiple of the identity. Combined with submultiplicativity and $\| \Psi_\theta \| = 1$, this gives the rate lock as an unavoidable algebraic consequence.

Affine channel. The dyadic content is carried not by the rate but by the *location* of the joint fixed point. Every nonzero bias vector b_A breaks G_2-equivariance: the only G_2-fixed vector in the irreducible adjoint is $b_A = 0$, so $b_A \neq 0$ is the symmetry-breaking order parameter that encodes the observer's identity. The coupling angle θ rotates the two observers' bias vectors into a joint consensus:

- **Theorem 9.4 (Affine consensus).** For any G_2-structured dyad (affine maps T_A, T_B with orthogonal orientations R_A, R_B and bias vectors b_A, b_B), the joint fixed point $(\hat{x}(\theta), \hat{y}(\theta))$ exists, is unique, and depends real-analytically on θ, with a matrix-valued closed form at $\theta = 0$ and $\theta = \pi/2$.

- **Proposition 9.5 (Conditional min-coherence gain).** Under the minimum-per-observer coherence functional $C_{min}(\theta) = min(C_A(\theta), C_B(\theta))$, there exists a non-empty improvement region R in the space of (θ, φ) pairs (where $\varphi = \angle(b_A, b_B)$ is the relative bias-alignment angle), bounded away from the opposed-bias boundary $\varphi = \pi$ in the verified seeded geometry, with an optimal $\theta^{¿}$ that depends on the specific rotation geometry of R_A, R_B and is generically not at $\theta = \pi/4$. A 50-seed Monte Carlo (§5.6, Appendix V.3.i) corroborates the existence of a non-empty improvement region and the $\theta^{¿} \neq \pi/4$ property across all sampled seeds, while showing that the bounded-away-from-π property is geometry-dependent and holds in $\sim 64\%$ of seeds (32 of 50).

A supporting **Numerical Observation 5.5.1** records the asymmetric coherence response under opposed biases for the seeded geometry, which (together with Remark 5.5.3) motivates the choice of C_{min} over C_{avg} as the joint coherence functional.

1.2 What this paper does and does not claim

This paper claims:

-

- (a) Linear coupling between G_2-structured self-modeling observers cannot improve the joint convergence rate (Theorems 9.1, 9.2; Corollary 9.3). The rate channel is closed by Schur's lemma.
- (b) Under affine observers $T(x) = rRx + b$ with nonzero biases, the joint fixed point depends nontrivially on the coupling angle (Theorem 9.4), with an explicit closed form.
- (c) Under the minimum-per-observer coherence functional, a non-empty conditional-gain region exists in the space of coupling angles and bias-alignment angles (Proposition 9.5), bounded away from the opposed-bias boundary in the seeded verified geometry. A 50-seed Monte Carlo (§5.6, Appendix V.3.i) confirms a non-empty improvement region in $50/50$ sampled (R_A, R_B) pairs and a non-universal $\theta^{¿}$ in all 50, while showing the bounded-away-from-π property holds in $\sim 64\%$ of seeds and is therefore not asserted in the amended Conjecture 9.5′ (§5.6).

This paper does not claim:

- (a) That every coupling improves joint coherence — the gain is *conditional* on non-destructive bias geometry, and a substantial region of (θ, φ)-space exhibits coherence *degradation* rather than gain (see Appendix V.3 Task 5.4).
- (b) A generalization to $n \geq 3$ observers. The $\binom{n}{2}$-pairwise-angle structure is open (§10.1).
- (c) That current biological or artificial systems satisfy the full G_2 fixed-point condition required by the framework. The framework is silent on which realizations qualify (§8).
- (d) A connection to specific consciousness, attachment, or coherence theories outside the PCI/PME series. The claims are structural.
- (e) Any of the claims retired during drafting. An earlier internal draft proposed a strict speedup theorem, a saturating coherence-bonus formula, and a linear severance threshold; all three were retired in response to numerical verification and representation-theoretic analysis. The complete retraction record, with what refuted each claim and what replaced it, is tabulated in Appendix V.0.1.

1.3 Roadmap

§2 recapitulates the single-observer results of Paper 7 in a compact form sufficient for the product-space construction. §3 defines the joint $M_A \times M_B$ space, the block-rotation coupling Ψ_θ

(Lemma 3.3.1), and the diagonal action of the Paper-7 symmetries on the product. §4 proves the rate-channel triple (Theorems 9.1, 9.2, Corollary 9.3). §5 proves the affine-channel results (Lemma 5.2.1 rate-lock extension, Theorem 9.4 consensus, Numerical Observation 5.5.1 and Proposition 9.5 conditional min-coherence gain, Note 9.6 deferring rate-improvement to the nonlinear sequel). §6 discusses empirical accessibility of the min-coherence prediction and its operationalization in inter-brain and human-AI coupling settings, including Figure 1 (the ΔC_{avg} and ΔC_{min} heatmaps). §7 treats the joint-fixed-point location and geometry-dependent optima. §8 addresses the human-AI dyad as a conditional special case. §9 enumerates what the framework does not establish. §10 discusses connections to Papers 7, 8, 10 of this series and flags open problems. Appendix V contains the complete numerical-verification record from Φ (three independent passes totaling 2217 test configurations), plus the retraction record for claims retired during drafting (V.0.1).

---

§2. Single-Observer Recap

This section states, without full proofs, the results of Paper 7 of this series that are required inputs to the dyadic construction. Full proofs are in [Graise 2026a, DOI 10.5281/zenodo.19773185]; we give only the statements and the minimum notation needed.

2.1 Observer as Banach fixed-point

Let B be a Banach space and $M \subseteq B$ a closed subset, equipped with the norm $\| \cdot \|_B$. An **observer** in the sense of Paper 7 is a pair (M, T) where $T : M \to M$ is a Banach contraction: there exists $r \in$ ¿ such that

$$\| T(x) - T(y) \|_B \leq r \| x - y \|_B \text{ for all } x, y \in M .$$

By the Banach fixed-point theorem, T has a unique fixed point $x^{¿} \in M$, and iterating T from any $x_0 \in M$ converges to $x^{¿}$ at rate r. The fixed point is interpreted as the observer's **self-model** — the stable representation it maintains of its own state, dynamics, and inputs.

For the present paper, we adopt the **affine** subclass of observers — those of the form

$$T(x) = r \cdot R \cdot x + b ,$$

with $r \in$ ¿ a scalar contraction rate, $R \in O(M)$ an orthogonal orientation operator, and $b \in M$ a bias vector. Such a map is a Banach contraction with rate r and fixed point $x^{¿} = (I - r R)^{-1} b$. We will see in §5.1 that the affine form is not an incidental simplification: it is the generic form of a *G_2-structured* self-modeling map, with $r R$ capturing the G_2-equivariant linear dynamics and b capturing the symmetry-breaking observer identity.

2.2 G_2 structure and the 14-dim adjoint representation

Paper 7 constrains the observer by requiring M to carry a G_2 structure in the sense of admitting a distinguished octonion-associative 3-form φ_{ijk} preserved by the compact Lie group G_2. The Lie algebra g_2 is 14-dimensional; we take the observer's Banach manifold M to be (an open region of) the 14-dim real vector space V^{14} carrying the adjoint representation of G_2. This adjoint representation is irreducible over R; equivalently, the commutant algebra of G_2 acting on V^{14} is $End_{G_2}(V^{14}) = R$, by real Schur's lemma.

The finite subgroup structure of G_2 includes $PSL(2,7)$ as the orientation stabilizer of φ, with $F_{21} = Z_7 \rtimes Z_3$ as the Fano-line-preserving subgroup (Paper 4, DOI 10.5281/zenodo.19617662). The self-modeling map T is required to be F_{21}-equivariant: $T(g \cdot x) = g \cdot T(x)$ for all $g \in F_{21}$.

2.3 The blind-spot bound and coherence ceiling

Paper 7 §3.4 proves that any F_{21}-equivariant Banach contraction on the 7-dim imaginary-octonion space $V^7 = Im(O)$ cannot capture the full self-model without an irreducible residual: the observer's fixed point on V^7 decomposes as $x^{¿} = x^{¿}_{accessible} + x^{¿}_{blind}$ with

$$\| x^{¿}_{blind} \|_B \geq \frac{1}{7} \cdot \| x^{¿} \|_B .$$

The factor $1/7$ is the **blind-spot ratio** ε_{min}; it arises as the ratio of the F_{21}-singlet dimension to the full 7-dim imaginary-octonion space. The complement

$$C_{max} = 1 - \varepsilon_{min} = \frac{6}{7}$$

is the **coherence ceiling**, the maximum stable value of the normalized coherence functional on V^7. The contraction rate r on V^7 is consequently bounded above by $6/7$: a contraction exceeding this rate would require access to the F_{21}-singlet direction, which the equivariance condition forbids.

Representation switch from Paper 7 (V^7) to Paper 9 (V^{14}). Paper 7's blind-spot theorem is stated on the 7-dim representation V^7 of G_2 (the imaginary octonions). In Paper 9 we work on the 14-dim **adjoint** representation V^{14}, where each observer is specified by an orthogonal transformation $R \in O(V^{14})$ and a bias $b \in V^{14}$ (§3.1). V^7 and V^{14} are *different* irreducible representations of G_2 — not subspaces of each other — so we are explicit about how the Paper 7 bound transfers, since this is a real shift of the underlying Banach space and not a notational rebadging.

The per-observer bound enters Paper 9 not as a relation about V^{14} itself but as a *constraint inherited from each observer's internal self-modeling subsystem*. Concretely, each observer in Paper 9 is assumed to satisfy the Paper 7 bound on its own V^7 self-modeling component: the contraction rate r_A and r_B used throughout this paper are bounded above by $6/7$ because each individual observer is a Paper 7 observer in its own right (§2.1, axiom inherited from Paper 7). The 14-dim space V^{14} is the **coupling/joint-state space** used to formalize the dyadic structure: it is where the orthogonal orientations R_A, R_B live, where the canonical block-rotation coupling Ψ_θ acts (Lemma 3.3.1), and where the consensus of Theorem 9.4 is computed. We do *not* claim a $1/14$ blind-spot ratio, a $13/14$ coherence ceiling, or any analogous Paper-7-style theorem on V^{14} in the present paper; those would require a separate F_{21}-branching analysis of the adjoint representation, which we leave as an open problem (§10.2). When we say in §5 that "$C_A, C_B \leq 6/7$", we mean each observer's *internal self-coherence on its own* V^7

subsystem is so bounded by Paper 7; the Paper 9 quantities $C_1(\theta)=|\langle \hat{x}, b_A \rangle|/\|\hat{x}\|$ and $C_2(\theta)$ measure cosine alignments on V^{14} and are distinct objects whose values may exceed or fall below $6/7$ without violating Paper 7.

This division of labor is the correct way to read the present paper: *Paper 7's bound is a per-observer constraint on each observer's internal V^7 blind-spot structure*; *Paper 9 builds dyadic structure on V^{14} where the coupling lives*. The two are linked through the common G_2 symmetry group but not through a shared underlying vector space.

2.4 The branching ratio prediction

Applying a fluctuation-dissipation-theorem (FDT) argument in Paper 7 §4.2 yields the predicted branching ratio for the neural-avalanche regime:

$$\sigma_{pred}=1-\frac{1}{49}=1-\varepsilon_{min}^2\approx 0.9796,$$

with the $1/49=(1/7)^2$ factor entering squared because the branching-ratio observable is quadratic in the fluctuation amplitude. Paper 7 §5 compares this prediction to the Wilting–Priesemann MR-estimator value $\hat{\sigma}\approx 0.98$ in awake mammalian cortex [Wilting & Priesemann 2018, DOI 10.1038/s41467-018-04725-4].

2.5 What the single-observer framework leaves open

Paper 7 treats a single observer with a single self-modeling map T on a single G_2-structured 14-dim manifold. It does not address:

1. **Interactions between multiple observers.** When two observers share information, mutually model each other, or couple their dynamics through a shared structure, the single-T formalism is inadequate.
2. **Emergent coherence from coupling.** If two single-observer ceilings are $6/7$ each, the naive expectation is that a coupled pair cannot exceed $6/7$ — any "group coherence" would still be bounded by the individual ceilings. Paper 7 does not address whether this naive expectation holds, nor whether coupling can change the *location* of the joint fixed point even when the *rate* is bounded.
3. **Observer identity as an algebraic object.** The symmetry-breaking role of the affine bias b — which distinguishes one observer's target self-model from another's — is implicit but not developed in Paper 7, which treats a single observer where there is nothing for b to be distinguished from.

The remainder of this paper addresses all three. The rate question (item 2, partial) is closed in the negative in §4 (Theorems 9.1, 9.2; Corollary 9.3): no linear coupling improves the joint rate. The joint-fixed-point question (item 2, full) is opened in §5 (Theorem 9.4, Proposition 9.5): coupling *does* change the fixed-point location, and under the right functional this produces conditional coherence gain. The identity question (item 3) is answered in §5.1: the affine bias vector b is the G_2-symmetry-breaking order parameter of each observer, and the dyadic consensus of Theorem 9.4 is a rotation-weighted combination of the two observers' biases.

§3. The Product-Space Construction

This section fixes the mathematical setting for the remainder of the paper. We construct the product Banach manifold on which the dyadic self-modeling dynamics will act, identify the canonical coupling map (an antisymmetric block rotation), and verify its three structural properties: isometry, one-parameter-group composition, and G_2-equivariance under the diagonal action.

3.1 Product observers

Let (M_A, T_A) and (M_B, T_B) be two G_2-structured Banach observers in the sense of §2.1, with $M_A, M_B \cong V^{14}$ each carrying the irreducible 14-dim adjoint representation of G_2. By §2.1, each observer is affine:

$$T_A(x) = r_A R_A x + b_A, T_B(y) = r_B R_B y + b_B,$$

with $r_A, r_B \in [0, 6/7)$, $R_A, R_B \in O(V^{14})$ orthogonal orientations, and $b_A, b_B \in V^{14}$ bias vectors.

We form the product

$$M_{AB} = M_A \times M_B \cong V^{14} \oplus V^{14},$$

equipped with the **product norm**

$$\|(x, y)\|_{AB} = \sqrt{\|x\|_A^2 + \|y\|_B^2}.$$

This makes M_{AB} a Banach subset of the direct sum $B_A \oplus B_B$ of total real dimension 28.

If the two observers were fully independent, the natural self-modeling map on the product would be the direct product $T_A \times T_B : (x, y) \mapsto (T_A(x), T_B(y))$, with joint contraction rate $max(r_A, r_B)$ and joint fixed point $(x_A^{¿}, y_B^{¿})$. No coupling would occur, no shared coherence would develop, and the dyad would factor trivially. This is not the object of the present paper.

3.2 The coupling map

To model inter-observer interaction, we introduce a bounded linear map

$$\Psi : M_{AB} \to M_{AB}$$

that we call the **coupling map**. The joint self-modeling map is then

$$T_{AB} = (T_A \times T_B) \circ \Psi : M_{AB} \to M_{AB}.$$

When $\Psi = id$, the joint map reduces to the decoupled product $T_A \times T_B$. When Ψ has nontrivial structure, the two observers' states are mixed before each self-modeling step, and the joint fixed point of T_{AB} need not factor as $(x_A^{¿}, y_B^{¿})$.

The choice of Ψ is not free: §4 will show that natural-looking choices can be either *amplifying* (joint rate exceeds the worse of the two individual rates — Theorem 9.1) or *rate-inert* (joint rate equals the worse individual rate — Theorem 9.2), with no choice that produces strict speedup. We therefore commit to a single canonical choice and prove its structural properties.

3.3 The block-rotation coupling and Lemma 3.3.1

The **canonical coupling** of this paper is the antisymmetric block rotation, parameterized by a single angle $\theta \in [0, \pi/2)$:

$$\Psi_\theta = \begin{pmatrix} \cos\theta \cdot I_{14} & -\sin\theta \cdot I_{14} \\ \sin\theta \cdot I_{14} & \cos\theta \cdot I_{14} \end{pmatrix} \in SO(M_{AB}),$$

acting as

$$\Psi_\theta(x, y) = (\cos\theta \cdot x - \sin\theta \cdot y, \sin\theta \cdot x + \cos\theta \cdot y).$$

The decoupled limit is $\theta = 0$ ($\Psi_0 = id$); the maximally-coupled "swap-rotated" limit is $\theta = \pi/2$.

The reason for choosing this particular form — antisymmetric off-diagonal pattern, rather than a symmetric one — is captured by the following lemma, which is the structural backbone of the rate-channel results in §4 and the affine-channel proofs in §5.

Lemma 3.3.1 below (§3.3.1) establishes three properties of Ψ_θ that are the structural backbone of the rate-channel results of §4 and the affine-channel proofs of §5: (i) Ψ_θ is a product-norm isometry ($\|\Psi_\theta\|_{op} = 1$); (ii) the family $\{\Psi_\theta\}$ is a one-parameter group; (iii) Ψ_θ commutes with the diagonal G_2-action.

The crucial step in the isometry proof is the cancellation of the cross terms $\pm 2\sin\theta\cos\theta\langle x, y\rangle$ between the two factors, which arises because the off-diagonal blocks of Ψ_θ have opposite signs ($-\sin\theta$ in the upper-right, $+\sin\theta$ in the lower-left). This antisymmetric-off-diagonal pattern is what distinguishes Ψ_θ from the symmetric mixing $\Psi_{sym} = \alpha \cdot id + \beta \cdot P_{swap}$ explored in §4.1 (Theorem 9.1), where the cross terms add rather than cancel and the operator norm becomes $\alpha + \beta > 1$.

3.4 The diagonal G_2 action

Each individual observer carries the G_2 structure inherited from Paper 7. For the joint observer, we adopt the **diagonal action**:

$$g \cdot (x, y) = (g \cdot x, g \cdot y) \text{ for all } g \in G_2, (x, y) \in M_{AB}.$$

Both observers are acted on by the same group element simultaneously. This captures the physical intuition that a coupled dyad shares a common reference frame for the G_2 structure: if one observer rotates its associative-3-form labeling, the other rotates in step.

Alternative actions (independent action $(g, h) \cdot (x, y) = (g \cdot x, h \cdot y)$ or anti-diagonal action $g \cdot (x, y) = (g \cdot x, g^{-1} \cdot y)$) are not adopted. The reason is forced by Schur's lemma rather than by physical intuition alone, as we now spell out.

Schur-forced justification. A linear coupling $\Psi : V^{14} \oplus V^{14} \to V^{14} \oplus V^{14}$ that is required to be equivariant under a chosen G_2-action on the product must commute with every group element. Under the **diagonal** action $g \cdot (x, y) = (g \cdot x, g \cdot y)$, the product space decomposes as $V^{14} \oplus V^{14}$ where each summand is the same irreducible G_2-representation. By Schur's lemma $End_{G_2}(V^{14}) = R$, so the four blocks of Ψ are scalars; the equivariant couplings form a 4-parameter family $End_{G_2}(V^{14} \oplus V^{14}) \cong M_2(R)$, which contains the nontrivial isometric block-rotation family Ψ_θ of Lemma 3.3.1.

Under the **independent** action $(g, h) \cdot (x, y) = (g \cdot x, h \cdot y)$ with g, h allowed to vary independently in G_2, equivariance requires Ψ to commute with every pair (g, h). Writing Ψ in block form $\Psi = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, the intertwining conditions become $A g = g A$, $D h = h D$ (so A, D are G_2-equivariant on a single copy of V^{14}, hence scalars), and $B h = g B$, $C g = h C$ for all $g, h \in G_2$ independently. The latter force $B = C = 0$: taking g generic and $h = e$ gives $B = g B$ for all g, so B vanishes on the absence of trivial subrepresentations in V^{14}, i.e., $B = 0$; symmetrically $C = 0$. Equivariant couplings under the independent action therefore reduce to $\Psi = diag(\alpha I, \delta I)$ — the strictly **decoupled** system with no cross-talk. The independent action thus *forces* the dyad to be decoupled at the linear level and admits no nontrivial block-rotation coupling. Equivalently: only the diagonal action produces a 4-dimensional End-algebra on $V^{14} \oplus V^{14}$ containing off-diagonal blocks, and only this algebra contains the isometric block rotations of Lemma 3.3.1.

Under the **anti-diagonal** action $g \cdot (x, y) = (g \cdot x, g^{-1} \cdot y)$, equivariance requires $A g = g A$ (as before), $D g^{-1} = g^{-1} D$ (so D is G_2-equivariant, scalar), and the cross-block conditions $B g^{-1} = g B$ and $C g = g^{-1} C$, equivalent to B intertwining V^{14} with its conjugate representation. For real G_2 on real V^{14} the adjoint representation is self-dual, so a nonzero B is permitted in principle, but interpreting g^{-1} as the action on the second factor corresponds physically to a time-reversed or "anti-observer" partner; we do not adopt this convention because the dyads of interest in §6–§7 (synchronized inter-brain coupling, human–AI teaming) are co-temporal observers, not observer/anti-observer pairs.

We therefore adopt the diagonal action throughout: it is the unique choice (among the three natural candidates) that (i) admits a nontrivial G_2-equivariant linear coupling at all, (ii) gives the block-rotation family Ψ_θ as the canonical isometric coupling (Lemma 3.3.1), and (iii) preserves the physical interpretation of common-reference-frame G_2 structure.

By Lemma 3.3.1(iii), the canonical coupling Ψ_θ is G_2-equivariant under the diagonal action — every block of Ψ_θ is a scalar multiple of the identity on V^{14}, which commutes with every linear G_2-action.

3.5 The F_{21}-equivariance subgroup (inherited, not separately deployed)

Each individual observer in Paper 7 was required to be F_{21}-equivariant, where $F_{21} = Z_7 \rtimes Z_3$ is the Fano-line-preserving subgroup of $PSL(2,7) \subset G_2$ (Paper 4). Under the diagonal G_2-action of §3.4, the joint dyadic system inherits a diagonal F₂₁-action, and Lemma 3.3.1(iii) gives F₂₁-equivariance of Ψ_θ as a special case ($F_{21} \subset G_2$ and every block of Ψ_θ is G₂-equivariant).

Scope note. The F₂₁-equivariance structure is not used beyond this inheritance in the present paper. The §4 rate-channel results use only full G₂-equivariance (Schur on the irreducible 14-dim adjoint), and the §5 affine-channel results use only orthogonality of R_A, R_B (Lemma 5.2.1); neither relies on F₂₁ fixed vectors or F₂₁-branching structure on V^{14}. A natural question — whether the F₂₁-branching of V^{14} contains nonzero fixed vectors, which would constrain the bias direction b in a meaningful way and give the framework a genuine finite-symmetry invariant beyond G₂-equivariance — requires the branching tables of Paper 4 and is left as an open problem (§10.2).

3.6 Summary of §3

We have fixed the following mathematical setting for the remainder of the paper:

1. **Product space:** $M_{AB} = M_A \times M_B$, product Banach norm, total real dimension 28.
2. **Affine observer maps:** $T_A(x) = r_A R_A x + b_A$, $T_B(y) = r_B R_B y + b_B$, with $r_A, r_B \in [0, 6/7]$, $R_A, R_B \in O(V^{14})$, $b_A, b_B \in V^{14}$.
3. **Canonical coupling:** Ψ_θ block rotation, $\theta \in [0, \pi/2]$ (Lemma 3.3.1).
4. **Joint self-modeling map:** $T_{AB} = (T_A \times T_B) \circ \Psi_\theta$.
5. **Group action:** diagonal G_2-action and its F_{21}-restriction on the product.

Lemma 3.6.1 (Schur-derived uniqueness of Ψ_θ). *The one-parameter family $\{\Psi_\theta\}_{\theta \in ¿¿}$ is the unique connected family of linear couplings on $M_{AB} = V^{14} \oplus V^{14}$ satisfying: (a)* ***Isometry:*** *$\Psi_\theta^\dagger \Psi_\theta = I$; (b)* ***G₂-equivariance under the diagonal action:*** *$\Psi_\theta \circ \Delta(g) = \Delta(g) \circ \Psi_\theta$ for all $g \in G_2$; (c)* ***Identity at*** *$\theta = 0$: $\Psi_0 = I_{M_{AB}}$.*

Up to choice of orientation, this family is the block rotation $\Psi_\theta = \begin{pmatrix} \cos\theta\, I & -\sin\theta\, I \\ \sin\theta\, I & \cos\theta\, I \end{pmatrix}$.

Proof. By real Schur's lemma applied to the irreducible 14-dim G₂ adjoint representation, $End_{G_2}(V^{14}) = R$. A G₂-equivariant linear endomorphism of $V^{14} \oplus V^{14}$ under the diagonal G₂-action therefore has, in any decomposition into block operators on the two V^{14} summands, all four blocks acting as real scalar multiples of $I_{V^{14}}$. Such endomorphisms form the R-algebra $M_2(R)$ — a 4-dimensional real parameter space.

Imposing isometry (condition a) restricts $M_2(R)$ to the orthogonal subgroup $O(2)$ (orthogonal 2×2 matrices), a 1-dimensional manifold with two connected components. Imposing connectivity to $\Psi_0 = I$ (condition c) selects the connected component of identity, which is $SO(2)$. Every element of $SO(2)$ is of the form $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ for a unique $\theta \in ¿$. Tensoring with $I_{V^{14}}$ on each block recovers the block-rotation form of Ψ_θ stated in §3.3. ▫

Why this lemma matters. The block-rotation coupling is not a modeling choice that we picked from a larger zoo of options for aesthetic reasons. It is the unique connected family of linear couplings satisfying the three structural conditions (isometry, G_2-equivariance, identity at $\theta=0$). Other natural couplings (the symmetric-diagonal mixing Ψ_{sym} of §4.1) fail condition (a) and produce amplification rather than rate-preservation, as Theorem 9.1 quantifies. Lemma 3.6.1 thus closes the loop between §3's setup and §4's results: the structural framework forces the specific coupling, and Theorem 9.2 then characterizes its rate behavior.

§4 will establish the rate-channel triple (Theorems 9.1, 9.2, Corollary 9.3) using this setup. §5 will turn to the affine channel (Theorem 9.4, Numerical Observation 5.5.1, Proposition 9.5).

3.3.1 Three Properties

Lemma 3.3.1 (Block-rotation isometry). *The map Ψ_θ defined above has the following three properties for every $\theta \in R$:*

(i) ***Isometry on the product norm.*** *For every $(x, y) \in M_A \oplus M_B$,*

$$\| \Psi_\theta(x, y) \|_{AB}^2 = \| x \|_A^2 + \| y \|_B^2 .$$

Equivalently, $\Psi_\theta^\dagger \Psi_\theta = id_{M_{AB}}$ and $\| \Psi_\theta \|_{op} = 1$.

(ii) ***One-parameter group.*** *$\Psi_{\theta_1} \circ \Psi_{\theta_2} = \Psi_{\theta_1 + \theta_2}$; in particular $\Psi_0 = id$ and $\Psi_\theta^{-1} = \Psi_{-\theta}$.*

(iii) ***G_2-equivariance under the diagonal action.*** *For every $g \in G_2$ and every (x, y),*

$$\Psi_\theta(g \cdot x, g \cdot y) = (g \cdot (\cos\theta \cdot x - \sin\theta \cdot y), g \cdot (\sin\theta \cdot x + \cos\theta \cdot y)) .$$

That is, Ψ_θ commutes with the diagonal G_2-action $\Delta(g) = g \oplus g$ on M_{AB}.

Proof. We verify each claim by direct computation.

(i) Isometry. Expand the product-norm of the image:

$$\| \Psi_\theta(x, y) \|_{AB}^2 = \| \cos\theta \cdot x - \sin\theta \cdot y \|_A^2 + \| \sin\theta \cdot x + \cos\theta \cdot y \|_B^2 .$$

Using the identification $\| \cdot \|_A = \| \cdot \|_B$ (same norm inherited from the representation):

$$¿ \cos^2\theta \| x \|^2 + \sin^2\theta \| y \|^2 - 2 \sin\theta \cos\theta \langle x, y \rangle$$

$$+ \sin^2\theta \| x \|^2 + \cos^2\theta \| y \|^2 + 2 \sin\theta \cos\theta \langle x, y \rangle .$$

The cross terms $-2 \sin\theta \cos\theta \langle x, y \rangle$ and $+2 \sin\theta \cos\theta \langle x, y \rangle$ cancel exactly (the key antisymmetric feature that distinguishes Ψ_θ from the symmetric mixing of Theorem 9.1). The diagonal terms sum to $(\cos^2\theta + \sin^2\theta)(\| x \|^2 + \| y \|^2) = \| x \|^2 + \| y \|^2$. Therefore $\| \Psi_\theta(x, y) \|_{AB}^2 = \| x \|^2 + \| y \|^2$.

Equivalently, $\Psi_\theta^\dagger \Psi_\theta = I$ (direct block-matrix computation: $R^\dagger R = I$ for any rotation R), and the spectral norm of a rotation is 1. ▫ (i)

(ii) One-parameter group. By direct composition of the block matrices:

$$\Psi_{\theta_1} \Psi_{\theta_2} = \begin{pmatrix} c_1 I & -s_1 I \\ s_1 I & c_1 I \end{pmatrix} \begin{pmatrix} c_2 I & -s_2 I \\ s_2 I & c_2 I \end{pmatrix}$$

$$\text{¿} \begin{pmatrix} (c_1 c_2 - s_1 s_2) I & -(c_1 s_2 + s_1 c_2) I \\ (c_1 s_2 + s_1 c_2) I & (c_1 c_2 - s_1 s_2) I \end{pmatrix}$$

$$\text{¿} \begin{pmatrix} \cos(\theta_1+\theta_2) I & -\sin(\theta_1+\theta_2) I \\ \sin(\theta_1+\theta_2) I & \cos(\theta_1+\theta_2) I \end{pmatrix} = \Psi_{\theta_1+\theta_2}.$$

Consequently $\Psi_0 = I$ and $\Psi_\theta^{-1} = \Psi_{-\theta}$. ▫ (ii)

(iii) G₂-equivariance. Each block of Ψ_θ is a real scalar multiple of the identity I_{14} on M_A (respectively, M_B) under the canonical identification. The scalar identity commutes with any linear action, in particular with the G_2-action on the representation. The $-\sin\theta$ and $+\sin\theta$ off-diagonal blocks likewise act as scalars times the canonical identification J_{AB}, which by construction intertwines the two G_2-actions. Hence every block of Ψ_θ is G_2-equivariant, and so is Ψ_θ itself.

Formally: for any $g \in G_2$,

$$\Psi_\theta \circ (g \oplus g) = \begin{pmatrix} cI & -sI \\ sI & cI \end{pmatrix} \begin{pmatrix} g & 0 \\ 0 & g \end{pmatrix} = \begin{pmatrix} cg & -sg \\ sg & cg \end{pmatrix} = \begin{pmatrix} g & 0 \\ 0 & g \end{pmatrix} \begin{pmatrix} cI & -sI \\ sI & cI \end{pmatrix} = (g \oplus g) \circ \Psi_\theta,$$

where the central equality uses that cI and sI commute with g (scalars commute with every linear map). ▫ (iii)

This completes the proof of Lemma 3.3.1. ▰

3.3.2 Remark — What Distinguishes Ψ_θ from the Symmetric Mix

The v0.9 draft of this paper used the **symmetric-diagonal coupling** $\Psi_{sym} = \alpha \cdot id + \beta \cdot P_{swap}$, with $\alpha^2 + \beta^2 = 1$, and incorrectly claimed that this map was an isometry. The spectral structure of Ψ_{sym} is: - symmetric eigendirection (x, x): eigenvalue $\alpha + \beta$, - antisymmetric eigendirection $(x, -x)$: eigenvalue $\alpha - \beta$,

so $\|\Psi_{sym}\|_{op} = \alpha + \beta > 1$ whenever $\beta > 0$. The symmetric mixing **amplifies** on the symmetric eigendirection, which propagates into Theorem 9.1's no-go result.

The block rotation Ψ_θ differs in exactly one sign: the off-diagonal block in the lower-left row carries $+\sin\theta$ while the upper-right block carries $-\sin\theta$ (antisymmetric off-diagonal pattern). The cancellation of cross terms in part (i) of the proof is a direct consequence of this antisymmetry, and it is what forces the eigenvalues onto the unit circle. The block rotation is the unique two-parameter family of coupling maps that is (a) isometric, (b) G_2-equivariant, and (c) reduces to the identity at $\theta=0$, up to a choice of orientation.

---

§4. The Rate Channel: Two Negative Results and Schur's Lemma

This section establishes three results about the **rate channel** of dyadic coupling. They are the structural opening of the paper. None of them claims that coupling improves convergence; together they prove that, in the linear G_2-equivariant setting, no such improvement is possible.

The naming is deliberate. The original program of this paper sought a strict speed-up theorem. Numerical investigation at 50-digit precision (Appendix V) refuted that program twice: once by exhibiting a configuration in which coupling makes things *worse* (Theorem 9.1 below), and once by exhibiting a configuration in which coupling has *no effect* on the rate (Theorem 9.2). Corollary 9.3 explains why no third configuration is going to rescue the speed-up: the obstruction is Schur's lemma applied to the irreducible 14-dimensional adjoint representation of G_2. The rate channel is closed.

The positive content of the paper — the **affine consensus theorem** — lives in §5, which works in the symmetry-broken category that the rate-lock of §4 forces us into. There is more to dyadic coupling than rate, and §4 isolates which channel carries it.

4.1 Theorem 9.1 — Symmetric coupling amplifies

The first natural attempt at a coupling map is the symmetric-diagonal mix $\Psi_{sym}(x,y)=(\alpha x+\beta y,\beta x+\alpha y)$, with $\alpha^2+\beta^2=1$. This appears in early drafts of the dyadic construction and has the correct invariance under observer-swap. It is not, however, an isometry of the product Banach space.

Theorem 9.1 (Symmetric coupling amplifies). *Let $\Psi_{sym}=\alpha\cdot id+\beta\cdot P_{swap}$ on $M_A\times M_B$, with $\alpha^2+\beta^2=1$ and $\alpha,\beta>0$ (so both blocks are non-trivial). Then*

$$\|\Psi_{sym}\|_{op}=\alpha+\beta>1.$$

Consequently, for $T_A=r_A I$ and $T_B=r_B I$ scalar contractions ($r_A,r_B\in(0,1)$), the joint map $T_{AB}=(T_A\oplus T_B)\circ\Psi_{sym}$ has worst-case rate

$$r_{AB}^{sym}\geq r_-\cdot(\alpha+\beta)>r_-,$$

where $r_-=\min(r_A,r_B)$, with equality on the symmetric eigendirection of Ψ_{sym} when $r_A=r_B$.

Proof. The operator-norm claim is direct: Ψ_{sym} has eigendirections $(x,x)\in M_{AB}$ (symmetric) and $(x,-x)$ (antisymmetric), with eigenvalues $\alpha+\beta$ and $\alpha-\beta$ respectively. For $\alpha,\beta>0$, $\alpha+\beta>1$, so $\|\Psi_{sym}\|_{op}=\alpha+\beta$.

For the rate bound, we exhibit an input vector achieving the lower bound for *both* $r_A = r_B$ and $r_A \neq r_B$. Take $v = (x, x) \in M_{AB}$ with $x \in V^{14}$ unit. Then $\| v \|_{AB}^2 = 2$, $\Psi_{sym}(v) = (\alpha+\beta)(x, x)$, and

$$T_{AB}(v) = (T_A \oplus T_B)((\alpha+\beta)x, (\alpha+\beta)x) = (\alpha+\beta)(r_A x, r_B x).$$

Its norm is

$$\| T_{AB}(v) \|_{AB}^2 = (\alpha+\beta)^2 (r_A^2 + r_B^2),$$

so the operator-norm Rayleigh quotient on this input is

$$\frac{\| T_{AB}(v) \|_{AB}}{\| v \|_{AB}} = (\alpha+\beta) \sqrt{r_A^2 + r_B^2 / 2}.$$

Since $\sqrt{(r_A^2 + r_B^2)/2} \geq r_- = min(r_A, r_B)$ (with equality iff $r_A = r_B$), we obtain

$$r_{AB}^{sym} \geq \frac{\| T_{AB}(v) \|_{AB}}{\| v \|_{AB}} = (\alpha+\beta) \sqrt{r_A^2 + r_B^2 / 2} \geq (\alpha+\beta) r_- > r_-,$$

the last strict inequality from $\alpha + \beta > 1$. The bound holds for all r_A, r_B, with equality $r_{AB}^{sym} = r(\alpha+\beta)$ in the symmetric case $r_A = r_B = r$ (where the symmetric eigendirection is a true invariant subspace of $T_A \oplus T_B$). Numerical verification across a 24-configuration grid yields 0/24 PASS for the naïve bound $r_{AB} \leq max(r_A, r_B)$ in the v0.9 form (Appendix V, Φ Task 1). ▫

Interpretation. Coupling that mixes the two observer states symmetrically along their inner-product diagonal has an amplifying mode. This mode is the symmetric subspace, where the two observers' updates *constructively interfere* and overshoot the contraction. Far from improving joint convergence, this coupling **destabilizes** it. The lesson is that not every coupling helps; some are actively harmful.

This is a useful negative result to state explicitly, because the symmetric-diagonal coupling is the form most readers will reach for first. Theorem 9.1 forecloses it.

4.2 Theorem 9.2 — Isometric coupling is rate-inert

The natural correction to Ψ_{sym} is to demand that the coupling be a genuine isometry, $\| \Psi \|_{op} = 1$. The minimal such replacement is the antisymmetric block rotation

$$\Psi_\theta(x, y) = (\cos\theta \cdot x - \sin\theta \cdot y, \sin\theta \cdot x + \cos\theta \cdot y),$$

parameterized by $\theta \in [0, \pi/2)$. This is an element of the special orthogonal subgroup of $O(M_{AB})$ generated by the canonical identification $J_{AB} : M_A \to M_B$, and is G_2-equivariant for the diagonal G_2-action (Lemma 3.3.1).

Theorem 9.2 (Isometric coupling is rate-inert). *Let $T_A = r_A I_{14}$, $T_B = r_B I_{14}$ be scalar G₂-equivariant contractions on the irreducible 14-dim adjoint representation, with $r_A, r_B \in [0, 6/7)$. Let Ψ_θ be the block-rotation coupling above, and let $T_{AB} = (T_A \oplus T_B) \circ \Psi_\theta$. Then*

$$r_{AB} = max(r_A, r_B), \text{independent of } \theta.$$

The bound is tight for every $\theta \in [0, \pi/2)$.

Proof. Submultiplicativity of the operator norm gives

$$r_{AB} \leq \| T_A \oplus T_B \|_{op} \cdot \| \Psi_\theta \|_{op} = max(r_A, r_B) \cdot 1 = max(r_A, r_B).$$

For tightness, observe that singular values are invariant under orthogonal multiplication on the right: for any A and any orthogonal Ψ_θ, $\sigma_{max}(A \circ \Psi_\theta) = \sigma_{max}(A)$ because $(A\Psi_\theta)^\dagger (A\Psi_\theta) = \Psi_\theta^\dagger A^\dagger A \Psi_\theta$ is orthogonally conjugate to $A^\dagger A$ and has the same spectrum. Applied with $A = T_A \oplus T_B = r_A I \oplus r_B I$, we get $\sigma_{max}(T_{AB}) = \sigma_{max}(r_A I \oplus r_B I) = max(r_A, r_B)$ exactly. The maximizing input is $\Psi_\theta^{-1}(v_0)$ for any leading singular vector v_0 of $r_A I \oplus r_B I$ (e.g., $(x_0, 0)$ when $r_A \geq r_B$). Numerical confirmation across a 96-configuration grid at 50-digit precision (Appendix V, Φ Task 4.2: 24/24 PASS at $r_A = r_B$, 72/72 saturation at $r_A \neq r_B$, all to deviation ¿7×10^{-16}). ▫

Interpretation. Replacing the amplifying coupling with an isometric one removes the destabilization of Theorem 9.1, but it does not produce a speed-up. The rate is now exactly $max(r_A, r_B)$ — the worst observer's rate — for every coupling angle. Isometric coupling is **rate-inert**: it does not change how fast trajectories collapse to the joint fixed point.

This is the key negative result of the paper. The natural rescue of Theorem 9.1 fails to rescue the speed-up program; it merely moves us from "strictly worse" to "exactly the same." A reader expecting a speed-up theorem at this point should adjust expectations: §5 will show that the content of the coupling lives elsewhere.

4.3 Corollary 9.3 — Schur's lemma closes the rate channel

Theorems 9.1 and 9.2 are not isolated facts. They are two instances of a single algebraic obstruction: the irreducibility of the G₂ adjoint representation forces the linear part of every G₂-equivariant contraction to be a scalar.

Corollary 9.3 (Schur rate lock). *Let V be a finite-dimensional nontrivial irreducible real representation of G_2. Let $T : V \to V$ be a linear G₂-equivariant map. Then $T = r \cdot id_V$ for some $r \in R$.*

Proof. Real Schur's lemma: any G_2-equivariant endomorphism of an irreducible real representation is an element of the commutant algebra, which for the 7-dim and 14-dim irreducible representations of G_2 is R [Fulton–Harris 1991, §22; the relevant complex representations have $End_{G_2}(V)_C = C$, and the underlying real form has commutant R]. Therefore T is a real scalar multiple of the identity. ▫

Corollary to the Corollary. Combined with submultiplicativity and $\|\Psi_\theta\| = 1$, this gives the rate lock $r_{AB} = max(r_A, r_B)$ as an unavoidable algebraic consequence: - *Schur's lemma:* forces T_A, T_B to be scalar — no anisotropy to mix. - *Submultiplicativity + isometric coupling:* gives $r_{AB} \le max(r_A, r_B)$. - *Scalar maps:* achieve the bound exactly on aligned inputs.

The "rate channel" of the dyadic system — the set of mechanisms by which coupling could in principle change the convergence rate — is therefore *closed* in the linear G_2-equivariant category. Whatever physical content dyadic coupling carries, it does not enter through this channel.

4.4 What is locked, what is free

Theorems 9.1, 9.2, and Corollary 9.3 collectively name what is locked. It is worth being equally explicit about what is free.

Locked: the linear, equivariant, rate channel. Schur's lemma forces $T = r\,I$. Submultiplicativity caps r_{AB} at $max(r_A, r_B)$. Isometric coupling achieves the cap exactly. There is no maneuver in this category that yields a strict speed-up.

Free: the affine, symmetry-breaking, target channel. The map $T_A(x) = r_A x + b_A$ with nonzero b_A is *not* fully G_2-equivariant. Strict G_2-equivariance of an affine map requires $b_A = g \cdot b_A$ for all $g \in G_2$, and the only such fixed vector in the irreducible 14-dim adjoint representation is $b_A = 0$. So the bias vector $b_A \neq 0$ **is the symmetry break**.

This is not a defect of the formalism. It is its content. The bias vector b_A is the order parameter of the observer: the direction in the 14-dimensional model space along which T_A is *not* G_2-symmetric. It is the location in the representation that this particular observer "selects" as its target. Two distinct observers have two distinct bias vectors; the dyadic coupling mixes them.

The affine bias channel is *not* governed by Schur's lemma. It is a 14-dimensional vector space of choices, one b per observer, with no algebraic restriction beyond the contraction condition $|r_A| < 1$. The Banach fixed-point of the joint affine map depends on θ through exactly this bias data; the rate does not.

The geometric picture:

- **Linear part** $(r_A I, r_B I)$: an isotropic shrinkage of the model space toward the origin. G_2-equivariant. Rate-determining. Locked by Schur.
- **Affine bias** (b_A, b_B): a directional offset of the contraction center. G_2-symmetry-breaking. Fixed-point-determining. Free.
- **Coupling angle** (θ): the rotation that mixes the two bias vectors. Determines the joint fixed point's location.

The thesis of the paper is that **dyadic coherence lives in the affine channel, not the rate channel**. §5 makes this precise via the affine consensus theorem (Theorem 9.4) and the conditional coherence-gain proposition (Proposition 9.5).

4.5 Connection to Paper 7

Theorem 9.2 strengthens rather than contradicts Paper 7's $6/7$ rate bound. Paper 7 establishes that any single G_2-equivariant self-modeling observer has contraction rate $\leq 6/7$. Theorem 9.2 establishes that any linearly coupled dyad of such observers retains the rate $max(r_A, r_B) \leq 6/7$. The dyadic case inherits the single-observer ceiling without improvement.

This is the consistent extension of Paper 7 to the dyadic setting: the $6/7$ bound is robust under linear isometric coupling.

The present forms of Theorems 9.1 and 9.2 are the corrected versions that Φ's numerical verification (Appendix V.1, V.2) supports without exception; the retraction record of earlier claims is consolidated in Appendix V.0.1.

---

§5. The Affine Channel: Consensus and Conditional Coherence Gain

§4 established that the linear, G_2-equivariant *rate channel* is closed by Schur's lemma and submultiplicativity. This section identifies the **affine channel** as the locus of dyadic content. Under affine contractions $T_A(x) = r_A R_A x + b_A$, $T_B(y) = r_B R_B y + b_B$ — where R_A, R_B are orthogonal endomorphisms of the model space and b_A, b_B are *symmetry-breaking* bias vectors that select each observer's target self-model — the joint fixed point depends nontrivially on the coupling angle θ, and its location traces a one-parameter family of consensus states. The rate is still locked at $max(r_A, r_B)$ (Theorem 9.2) but the *destination* of joint convergence is a function of θ.

The two main results: **Theorem 9.4** (affine consensus) establishes existence, uniqueness, and a closed form for the joint fixed point. **Proposition 9.5** (conditional coherence gain) characterizes the geometric conditions under which the joint state's *minimum* per-observer coherence exceeds the decoupled minimum. The minimum functional, rather than the average, is the correct measure: a structural finding from the verification (Φ Task 5.6) showed that average coherence can register illusory gains in opposed-bias geometries due to one observer accidentally aligning with the other's target while the first falls. Minimum coherence captures the honest "no observer gets worse" condition.

5.1 Affine observers and the scope of the G_2 structure

Where the G_2 structure does work in this paper. The G_2 adjoint representation V^{14} and Schur's lemma are essential to §4: the rate-lock Theorem 9.2 and Corollary 9.3 require *G_2-equivariant* linear contractions on an *irreducible* G_2-module, and both hypotheses are used. The

§5 affine-channel results, by contrast, use only two features of V^{14}: (i) its real dimension (fixed at 14 by the choice of G_2 adjoint as the ambient space from Paper 7) and (ii) the presence of a G_2-invariant Euclidean norm (so that the product norm of §3.1 and the isometry of Ψ_θ of Lemma 3.3.1 are well-defined). The affine observers themselves are *not* required to be G_2-equivariant: Lemma 5.2.1 establishes the joint-rate lock for any orthogonal $R_A, R_B \in O(V^{14})$, and Theorem 9.4 and Proposition 9.5 inherit the same orthogonal-only hypothesis.

This is deliberate. Remark 5.5.2 (below) exhibits a counterexample: when $R_A = R_B = I$ (the strictly G_2-equivariant case forced by Schur when both linear parts commute with G_2), the affine channel *collapses* as well — $C_1(\theta) = 1$ identically and the conditional-gain structure of Proposition 9.5 is trivial. The interesting regime, where coupling changes the joint fixed point in a nontrivial geometry-dependent way, requires R_A and R_B to *not* commute with the G_2-action. The §5 results are therefore best read as results about *general affine Banach dyads on a 14-dim Euclidean space*, obtained by relaxing the G_2-equivariance hypothesis of §4.

This scope-relaxation is not a weakness of the paper; it is the content of the rate-channel-vs-affine-channel dichotomy. §4 characterizes what is locked when G_2-equivariance is enforced on the linear part; §5 characterizes what opens when it is relaxed to orthogonality. The bias $b \neq 0$ is separately a G_2-symmetry-breaking order parameter (Schur forbids nonzero G_2-fixed vectors in the irreducible adjoint), but the main channel that relaxes G_2 is the orthogonal-but-not-equivariant linear part R — the bias enters the joint dynamics through R, not through any independent G_2 structure.

Terminology (clarified). Throughout this paper, - *G_2-equivariant* means: commutes with the G_2-action on V^{14} (i.e., in $Comm(G_2)$, which by real Schur is R). - *G_2-structured* means: acts on V^{14} equipped with the G_2-invariant Euclidean norm. This is a dimension-and-metric choice, not an equivariance condition.

The §4 rate-channel results require G_2-equivariant linear parts. The §5 affine-channel results require only G_2-structure in the metric sense.

A **G_2-structured affine observer** is a self-modeling map of the form

$$T(x) = r\,R\,x + b,$$

where: - $r \in [0, 6/7)$ is the **contraction rate** (a real positive scalar); - $R \in O(V^{14})$ is an **orientation operator** (orthogonal on the G_2-invariant Euclidean norm; may or may not commute with the G_2-action); - $b \in V^{14}$ is the **bias vector**, the symmetry-breaking order parameter that selects the observer's target self-model (and *cannot* be G_2-fixed unless $b = 0$).

The unique fixed point of T is $x^{¿} = (I - r\,R)^{-1} b$, which exists whenever $r\,R$ has spectral radius $r < 1$ — automatic for $r \in ¿$ since R is orthogonal and so $\| r\,R \|_{op} = r$. The fixed point lies in the affine span of b under iterated $r\,R$-action; for $R = I$ it reduces to the simple rescaling $x^{¿} = b/(1-r)$.

Why this form rather than $T(x) = r\,x + b$. The simpler scalar form ($R = I$) is a special case. Including a general orthogonal R allows the model to capture *internal model dynamics* that rotate within the representation space without changing its norm — a feature Φ's verification relied on, and which produces the §5.6 finding by exposing geometry-specific behavior that scalar models would have hidden.

Strict G₂-equivariance is broken by $b \neq 0$**.** Full G₂-equivariance of T would require both $R \in Comm(G_2)$ (G₂-equivariant orientation) *and* $g \cdot b = b$ for all $g \in G_2$. By Schur on the irreducible 14-dim adjoint, the only G₂-fixed vector is $b = 0$. Therefore *every nonzero observer breaks G₂-equivariance through its bias*, and that break is the signature of the observer's identity. The linear part of T may or may not commute with G₂; the affine part necessarily does not.

The **G₂-structured dyad** is a pair of such observers (T_A, T_B). Six parameters describe the linear data ($r_A, r_B \in [0, 6/7)$ and $R_A, R_B \in O(V^{14})$ each carrying $14 \cdot 13/2 = 91$ angles, modulo overall scale on the biases), plus two 14-vectors b_A, b_B and the coupling angle θ.

5.2 The joint affine map and its fixed-point equation

Coupling the dyad with the block rotation Ψ_θ of §3.3 (Lemma 3.3.1), the joint affine map on $M_{AB} \cong V^{14} \oplus V^{14}$ is

$$T_{AB}(x, y) = (r_A R_A(\cos\theta \cdot x - \sin\theta \cdot y) + b_A, r_B R_B(\sin\theta \cdot x + \cos\theta \cdot y) + b_B).$$

The fixed-point equation $T_{AB}(\hat{x}, \hat{y}) = (\hat{x}, \hat{y})$ is a 28×28 linear system:

$$M(\theta, r_A, R_A, r_B, R_B)\begin{pmatrix} \hat{x} \\ \hat{y} \end{pmatrix} = \begin{pmatrix} b_A \\ b_B \end{pmatrix}, M = \begin{pmatrix} I - r_A \cos\theta\, R_A & r_A \sin\theta\, R_A \\ -r_B \sin\theta\, R_B & I - r_B \cos\theta\, R_B \end{pmatrix}.$$

The block matrix M is invertible whenever the joint linear part $L_\theta = T_{AB} - b_{AB}$ has spectral radius less than 1. The next lemma establishes that $\| L_\theta \|_{op} = max(r_A, r_B)$ for arbitrary orthogonal R_A, R_B — not only the strictly G₂-equivariant case $R_A = R_B = I$ covered by Theorem 9.2.

Lemma 5.2.1 (Rate lock for orthogonal R_A, R_B**).** *For any orthogonal* $R_A, R_B \in O(V^{14})$*, any* $r_A, r_B \in$ ¿*, and any* $\theta \in [0, \pi/2]$*, the joint linear map* $L_\theta = (r_A R_A \oplus r_B R_B) \circ \Psi_\theta$ *has operator norm*

$$\| L_\theta \|_{op} = max(r_A, r_B).$$

Proof. The block-diagonal map $r_A R_A \oplus r_B R_B$ has operator norm $max(\| r_A R_A \|_{op}, \| r_B R_B \|_{op}) = max(r_A, r_B)$, because an orthogonal R has $\| R \|_{op} = 1$ so $\| r R \|_{op} = r$. The block rotation Ψ_θ is an orthogonal isometry by Lemma 3.3.1 with $\| \Psi_\theta \|_{op} = 1$. The operator norm is invariant under composition with an isometry on the right (singular values are preserved by orthogonal multiplication), so

$$\| L_\theta \|_{op} = \| (r_A R_A \oplus r_B R_B) \circ \Psi_\theta \|_{op} = \| r_A R_A \oplus r_B R_B \|_{op} = max(r_A, r_B). \square$$

The Schur-based Theorem 9.2 is the special case $R_A = R_B = I$ of Lemma 5.2.1; the difference is that Theorem 9.2 proves $r_A I, r_B I$ from Schur's lemma (forced by G₂-equivariance), while

Lemma 5.2.1 takes orthogonality of R_A, R_B as the input. The conclusion (rate lock) is the same. The affine channel of §5.3–5.7 takes advantage of the strictly weaker orthogonality assumption by allowing non-G₂-equivariant rotations — the framework's identity-encoding bias b_A requires a non-equivariant R_A to interact with the coupling θ in a non-trivial way (cf. Remark 5.5.2's $R_A = R_B = I$ counterexample).

With Lemma 5.2.1 in hand, $M = I - L_\theta$ is invertible for $r_A, r_B \in$ ¿ and any orthogonal R_A, R_B.

5.3 Theorem 9.4 — Affine Consensus

Theorem 9.4 (Affine Consensus). *Let $(T_A, T_B) = (r_A R_A x + b_A, r_B R_B y + b_B)$ be a G₂-structured dyad with $r_A, r_B \in [0, 6/7)$, $R_A, R_B \in O(V^{14})$, and $b_A, b_B \in V^{14}$. For each $\theta \in [0, \pi/2)$, the joint affine map T_{AB} has a unique fixed point $(\hat{x}(\theta), \hat{y}(\theta)) \in M_{AB}$, which varies real-analytically in θ.*

*(a) **Existence and uniqueness:** $\hat{x}, \hat{y}$ solve the 28×28 system $M \cdot (\hat{x}, \hat{y})^\top = (b_A, b_B)^\top$, where M is invertible because $\| L_\theta \|_{op} = max(r_A, r_B) < 1$ (Lemma 5.2.1 for general orthogonal R_A, R_B; Theorem 9.2 in the scalar specialization).*

*(b) **Reduction to individual fixed points at** $\theta = 0$**:** at the decoupled boundary,*

$$\hat{x}(0) = (I - r_A R_A)^{-1} b_A = x_A^{¿}, \hat{y}(0) = (I - r_B R_B)^{-1} b_B = y_B^{¿}.$$

*(c) **Closed form at** $\theta = \pi/2$ **(symmetric rates and orientation):** if $r_A = r_B = r$ and $R_A = R_B = R$, then*

$$\hat{x}(\pi/2) = (I + r^2 R^2)^{-1} (b_A - r R b_B), \hat{y}(\pi/2) = (I + r^2 R^2)^{-1} (r R b_A + b_B).$$

In particular, when $b_A = b_B = b$, using that R commutes with every polynomial in R so $(I + r^2 R^2)^{-1}$ commutes with $(I \pm r R)$:

$$\hat{x}(\pi/2) = (I - r R)(I + r^2 R^2)^{-1} b, \hat{y}(\pi/2) = (I + r R)(I + r^2 R^2)^{-1} b,$$

$$\hat{x} + \hat{y} = 2(I + r^2 R^2)^{-1} b, \hat{y} - \hat{x} = 2 r R (I + r^2 R^2)^{-1} b.$$

*(d) **Convergence rate** to the joint fixed point is $max(r_A, r_B)$, by Lemma 5.2.1 (for general orthogonal R_A, R_B) or Theorem 9.2 (for the strictly G₂-equivariant scalar specialization $R_A = R_B = I$).*

Proof. Part (a) by Banach: L_θ has operator norm $max(r_A, r_B) < 1$ (Theorem 9.2 and Lemma 5.2.1), so T_{AB} is a contraction with unique fixed point; equivalently, $M = I - L_\theta$ is invertible by the Neumann series, giving the explicit linear-system solution.

Part (b) by direct substitution: at $\theta = 0$, $M = diag(I - r_A R_A, I - r_B R_B)$, so the system decouples into two independent 14×14 inversions.

Part (c) by direct elimination. For $r_A = r_B = r$, $R_A = R_B = R$, and $\theta = \pi/2$ (so $\cos\theta = 0$, $\sin\theta = 1$), the joint map is

$$T_{AB}(x,y)=(-rRy+b_A, rRx+b_B),$$

and the fixed-point equations are

$$\hat{x}+rR\hat{y}=b_A, -rR\hat{x}+\hat{y}=b_B.$$

Solve by elimination. From the first equation, $\hat{x}=b_A-rR\hat{y}$. Substitute into the second:

$$-rR(b_A-rR\hat{y})+\hat{y}=b_B \Rightarrow (I+r^2R^2)\hat{y}=rRb_A+b_B$$

(using $-rR\cdot -rR=r^2R^2$). Invert to obtain $\hat{y}=(I+r^2R^2)^{-1}(rRb_A+b_B)$, then back-substitute:

$$\hat{x}=b_A-rR(I+r^2R^2)^{-1}(rRb_A+b_B)=(I+r^2R^2)^{-1}\left[(I+r^2R^2)b_A-rR(rRb_A+b_B)\right)$$

$$¿(I+r^2R^2)^{-1}\left[b_A+r^2R^2b_A-r^2R^2b_A-rRb_B\right)=(I+r^2R^2)^{-1}(b_A-rRb_B).$$

(The cancellation of $\pm r^2R^2b_A$ uses that R commutes with every polynomial in R.) This establishes the general formula. The symmetric $b_A=b_B=b$ specialization follows by substitution and by noting that R commutes with $(I+r^2R^2)^{-1}$, so $\hat{x}=(I+r^2R^2)^{-1}(I-rR)b=(I-rR)(I+r^2R^2)^{-1}b$. The sum and difference formulas follow by adding and subtracting. Numerical verification at 16-digit precision (Appendix V.3, Task 5.2 Check C) confirms the symmetric-biases specialization to deviation $¿7\times10^{-16}$; the general (distinct-biases) formula is verified in the accompanying review-pass counterexample test against the correct closed form above.

Part (d) follows from Lemma 5.2.1 (general orthogonal case) or, in the strictly G_2-equivariant scalar specialization, from Theorem 9.2. ▫

Real-analyticity. The map $\theta\mapsto M(\theta)^{-1}$ is real-analytic on the open set $\{\theta: det\,M(\theta)\neq 0\}=[0,\pi/2]$ (where $det\,M>0$ throughout), as a composition of real-analytic functions of $\sin\theta, \cos\theta$ and entry-wise matrix inversion. Hence $(\hat{x}(\theta), \hat{y}(\theta))$ depends real-analytically on θ.

Numerical verification (Appendix V.3, Task 5.1): for the test configuration $(r_A, r_B)=(0.7, 0.7)$ with R_A, R_B random orthogonal (seeds 20260504, 20260505) and b_A, b_B random unit (seeds 20260504, 20260554), $|\hat{x}|$ varies by 1.078 and $|\hat{x}-\hat{y}|$ by 0.406 across $\theta\in[0,\pi/2]$, both well above the nontriviality threshold. The minimum of $|\hat{x}-\hat{y}|$ occurs at $\theta=51.4°$, *not* at $\theta=45°$ or $\theta=90°$ — geometry-dependent rather than universal.

5.4 Linear-response expansion

For $\theta\ll1$, expand the closed form to leading order. Writing $M(\theta)=M(0)+\theta M'(0)+O(\theta^2)$ where $M(0)=diag(I-r_AR_A, I-r_BR_B)$ and $M'(0)=\begin{pmatrix}0 & r_AR_A\\ -r_BR_B & 0\end{pmatrix}$:

$$M(\theta)^{-1}=M(0)^{-1}-\theta M(0)^{-1}M'(0)M(0)^{-1}+O(\theta^2).$$

Applied to $(b_A, b_B)^\top$ (and using $M(0)^{-1} = diag\left((I - r_A R_A)^{-1}, (I - r_B R_B)^{-1}\right)$ applied to each side of $M'(0)$):

$$\hat{x}(\theta) = x_A^{¿} - \theta \cdot r_A (I - r_A R_A)^{-1} R_A \cdot y_B^{¿} + O(\theta^2),$$

$$\hat{y}(\theta) = y_B^{¿} + \theta \cdot r_B (I - r_B R_B)^{-1} R_B \cdot x_A^{¿} + O(\theta^2).$$

This is the **linear-response regime**: a small coupling angle θ pulls each observer's joint state by a fraction $r R (I - r R)^{-1}$ of *the other observer's individual fixed point*, modulated by the local rotation R. This is the formal weak-coupling perturbation that drives all of §5.5–§5.7. The sign asymmetry (minus for $\hat{x}$, plus for $\hat{y}$) is the signature of the antisymmetric off-diagonal pattern of Ψ_θ (§3.3).

5.5 Coherence and the choice of functional

To compare the joint fixed point with the individual fixed points, we need a coherence functional. Several natural choices exist; the distinction matters.

Per-observer coherence. For a nonzero vector v in the model space and a fixed unit reference direction e_{ref}, define

$$C_{ref}(v) = ¿\langle v / \| v \|, e_{ref} \rangle \vee \in [0, 1].$$

The absolute value ensures C takes values in $[0,1]$ rather than $[-1,1]$ — coherence is a measure of "alignment with the reference direction" independent of sign (orientation reversal is a gauge freedom in the self-model). For each observer in the dyad, the natural reference is the observer's own bias: $C_1 = C_{b_A}(\hat{x})$, $C_2 = C_{b_B}(\hat{y})$. These measure how self-aligned each observer's joint state is.

Joint-coherence aggregations. From the per-observer values, two natural aggregations:

$$C_{avg} = 1/2 (C_1 + C_2), C_{min} = min(C_1, C_2).$$

The choice between these matters. The *averaging functional* allows one observer's gain to mask the other's loss; under destructive geometry this produces illusory net improvement. The *minimum functional* tracks the worst-aligned observer and is monotonically degraded whenever any single observer's coherence falls.

This distinction is not academic. Φ's Task 5.6 verification (Appendix V.3) reports the following structural fact, which guides our choice of functional in Proposition 9.5 below.

Numerical Observation 5.5.1 (Asymmetric coherence response under opposed biases — verified geometry). *Fix the orthogonal pair* (R_A, R_B) *obtained from QR decomposition of standard-normal seeds 20260504 and 20260505 in dimension 14, take* $b_A = e_1$, $b_B = -e_1$, *and* $r_A = r_B = r = 0.7$. *Direct evaluation of the joint fixed point* $M(\theta) z = (b_A; b_B)$,

$M(\theta)=\begin{pmatrix} I-r\cos\theta\, R_A & r\sin\theta\, R_A \\ -r\sin\theta\, R_B & I-r\cos\theta\, R_B \end{pmatrix}$*, on a uniform sweep of* $\theta\in[0,\pi/2]$ *yields the following per-observer cosine alignments* $C_i(\theta)=|\langle \hat{x}_i(\theta), b_i\rangle|/\|\hat{x}_i(\theta)\|$*:*

θ (deg)	0	5	10	20	30	45	60	75	90
$C_1(\theta)$	0.7586	0.7313	0.7036	0.6543	0.6185	0.5906	0.5904	0.6148	0.6637
$C_2(\theta)$	0.7401	0.7691	0.8000	0.8600	0.9018	0.9123	0.8763	0.8228	0.7648
$C_{min}(\theta)$	0.7401	0.7313	0.7036	0.6543	0.6185	0.5906	0.5904	0.6148	0.6637

Two qualitative features are read off directly from the table and are stable under refinement of the θ*-sweep (see computations/paper9_C1C2_trajectory.py):*

(i) The two per-observer alignments respond with opposite sign to small θ *—* C_1 *falls (asymmetric blind-spot of observer A is magnified by the opposed coupling) while* C_2 *rises (the antipodal coupling lifts B's alignment with* $b_B=-b_A$ *via the rotation* $R_B\hat{x}$*).*

(ii) The min-aggregate $C_{min}(\theta)$ *is bounded above by its decoupled value* $C_{min}(0)=0.7401$ *on the entire sampled range* $\theta\in(0,\pi/2$¿*, with strict inequality for every sampled* $\theta>0$ *(minimum* ≈ 0.5875 *at* $\theta\approx 52°$*).*

Provenance. The numbers in the table are produced by computations/paper9_C1C2_trajectory.py, which constructs R_A, R_B from the recorded seeds (numpy.random.default_rng(20260504) and (20260505), then numpy.linalg.qr of a standard-normal 14 × 14 matrix, with NumPy's documented Givens-rotation orientation), solves the 28 × 28 linear system $M(\theta)z=(e_1;-e_1)$ at each tabulated θ, and reports the absolute cosine of the first 14- and last 14-coordinate blocks against $\pm e_1$ respectively. The script reproduces the table to four decimal places.

Status. This is a numerical observation about the seeded geometry, not a lemma. The signs of the two response coefficients $dC_1/d\theta$¿$_0$ and $dC_2/d\theta$¿$_0$ are observed (negative and positive, respectively) but we do not assert these signs hold for generic orthogonal (R_A, R_B); in particular $R_A=R_B=I$ gives $C_1(\theta)=C_2(\theta)=1$ identically. The structural use of the observation is recorded in Remark 5.5.2 below: it justifies our choice of C_{min} over C_{avg} for the dyadic functional, since the averaging functional would mask the strict C_1 degradation by the larger C_2 rise.

Remark 5.5.2 (No universality claim). Numerical Observation 5.5.1 is stated for a specific pair (R_A, R_B), namely the seeded pair used throughout the Φ verification. We do *not* claim that $dC_1/d\theta<0$ holds for generic orthogonal (R_A, R_B). In particular, the special case $R_A=R_B=I$ (the strictly G$_2$-equivariant case forced by Schur's lemma when both linear parts are equivariant) gives $dC_1/d\theta=0$ at $\theta=0$, and a direct calculation shows $C_1(\theta)=1$ identically along $\theta\in[0,\pi/2]$ when b_A is any nonzero vector (both $\hat{x}(\theta)$ and b_A point in the same one-dimensional subspace). The degradation observed numerically is a property of the *non-equivariant* rotation geometry of the seeded (R_A, R_B), not a universal fact about opposed biases.

Whether the degradation holds on a *generic* (Haar-typical) orthogonal pair (R_A, R_B) is open; we conjecture yes for R_A, R_B drawn from the Haar measure on $O(14)$¿$R:[R, b_A b_A^{\top}]=0\}$¿ (the complement of the codimension-thirteen subset that commutes with the b_A-rank-one projector), but a rigorous proof would require a measure- theoretic argument we do not provide. The Monte Carlo evidence of Appendix V.3 (50 seeded (R_A, R_B) pairs, all 50 exhibiting a non-empty C_{min} improvement region) supports the conjecture but does not prove it.

The purpose of Numerical Observation 5.5.1 in this paper is structural: it establishes that under the seeded geometry the *averaging* functional C_{avg} admits illusory gain (one observer rises while the other falls), so the *minimum* functional C_{min} is the honest dyadic aggregation. The motivation for C_{min} is derived in Remark 5.5.3 below from Paper 7's blind-spot framework, not from the numerical observation alone.

Remark 5.5.3 (Why C_{min} from first principles). Paper 7's single-observer result can be phrased as a *per-observer constraint*: for any G_2-structured self-modeling observer at its fixed point, the blind-spot ratio satisfies $\varepsilon \leq \varepsilon_{min} = 1/7$, equivalently $C \geq 1 - \varepsilon_{min} = 6/7$. The *honest dyadic extension* is that this per-observer constraint must hold for *each* observer in the dyad individually:

$$\varepsilon_A^{joint}(\theta) \leq \varepsilon_{min} \text{ and } \varepsilon_B^{joint}(\theta) \leq \varepsilon_{min},$$

equivalently $min(C_A(\theta), C_B(\theta)) \geq 1 - \varepsilon_{min}$.

The averaging functional $C_{avg} = 1/2(C_A + C_B)$ does *not* enforce this per-observer constraint: it permits one observer's blind-spot to exceed ε_{min} (i.e., its coherence to fall below $6/7$) if the other's is correspondingly elevated. From the per-observer Paper-7 standpoint, such trajectories violate the single-observer bound for at least one member of the dyad, and the aggregate functional registers them as acceptable by cancellation. The minimum functional is the unique linear-order-preserving aggregation that enforces the Paper 7 bound on *each* observer.

This is the principled reason we work with C_{min}, derived from the Paper 7 per-observer framework rather than chosen post hoc to make Proposition 9.5 hold cleanly.

5.6 Proposition 9.5 — Conditional Min-Coherence Gain

Proposition 9.5 (Conditional min-coherence gain — verified geometry). *Fix the seeded geometric configuration of Numerical Observation 5.5.1: R_A, R_B are the orthogonal matrices from QR of standard-normal seeds 20260504 and 20260505 in dimension 14; $r_A = r_B = r = 0.7$. Let $\varphi \in [0, \pi)$ be the bias-alignment angle* $\arccos(\langle b_A / \| b_A \|, b_B / \| b_B \| \rangle)$. *Let $C_{min}(\theta) = min(C_1(\theta), C_2(\theta))$ and $C_{min}^{dec} = C_{min}(0)$. Then:*

(a) ***Numerical existence of an improvement region for*** C_{min}*. Computing $C_{min}(\varphi, \theta)$ directly on the same 100 × 20 grid as Appendix V.3 Task 5.4 (re-evaluated with the $min(C_1, C_2)$ aggregation rather than the average), there exists a non-empty subset $\hat{R}_{min} \subset (\varphi, \theta) \in (0, \pi) \times (0, \pi/2)$ of positive Lebesgue measure on which $C_{min}(\theta) > C_{min}^{dec}$. For the seeded geometry, $\hat{R}_{min}$ occupies approximately 12.5% of the sampled grid (250 cells of 2000),*

strictly smaller than the corresponding C_{avg} improvement region (~25%), and bounded above by $\varphi = 176.4°$ (so strictly inside the opposed-bias boundary, consistent with part (b)).

(b) ***Degradation under opposed biases (verified geometry).*** *For the seeded (R_A, R_B) and $\varphi = \pi$ (opposed biases), the run of Appendix V.3 Task 5.6 finds $C_1(\theta) < C_1(0)$ for all $\theta \in (0, \pi/2¿$ tested. Since $C_{min} \le C_1$, $C_{min}(\theta) < C_{min}^{dec}$ on this θ-range under the seeded geometry.*

(c) ***No universal $\theta^¿$.*** *For the seeded geometry, the optimal coupling angle $\theta^¿ := \arg\max_\theta C_{min}(\theta)$ is $\approx 51°$ in the symmetric-rate case and $\approx 75° - 80°$ in the asymmetric-rate case (Appendix V.3 Tasks 5.1, 5.5), establishing that $\theta^¿$ is not generically at $\pi/4$.*

Proof. Part (a): direct from the C_{min} heatmap of Figure 1(b) and the supporting CSV (paper9_task5_heatmap_cmin.csv), computed by re-evaluating the joint fixed point on the same (φ, θ) grid as Task 5.4 and applying the $min(C_1, C_2)$ aggregation. The improvement region $\hat{R}_{min}$ is non-empty (250 cells of 2000) and spans $\varphi \in [18.2°, 176.4°]$, θ values concentrated in $[10°, 65°]$. Part (b): the C_1 trajectory of Appendix V.3 Task 5.6 falls from 0.7655 at $\theta = 0$ to 0.5955 at $\theta = \pi/2$ along the sampled grid; combined with $C_{min} \le C_1$ this gives the stated inequality. The C_{min} heatmap (Fig. 1(b)) confirms there are no positive cells at $\varphi = 180°$ for any sampled θ. Part (c): the optimal angles are read off from Tasks 5.1 and 5.5 grids directly. ▫

Conjecture 9.5′ (Conditional min-coherence gain — generic geometry). *For (R_A, R_B) drawn from a positive-measure subset of $O(14) \times O(14)$ (Haar measure), the qualitative content of Proposition 9.5 holds: a non-empty C_{min}-improvement region exists in (φ, θ)-space, and its optimal coupling angle $\theta^¿$ at aligned bias ($\varphi = 0$) is generically not at $\pi/4$. The region is not, in general, bounded strictly away from $\varphi = \pi$.*

We do not prove the conjecture. We have, however, conducted a Monte Carlo across 50 independently seeded (R_A, R_B) pairs (seeds $20260504, \ldots, 20260553$ for R_A and the next-incremented seed for R_B, each via QR of a standard-normal 14×14 matrix); the full per-seed CSV is computations/paper9_conjecture95_mc_seeds.csv, and the aggregate statistics are recorded in computations/paper9_conjecture95_mc_summary.json. The grid is the same 100×20 sweep over $(\varphi, \theta) \in [0, \pi) \times [0, \pi/2)$ used in Appendix V.3 Task 5.4, with bias $b_A = e_1$ and $b_B = \cos\varphi\, e_1 + \sin\varphi\, e_2$, $r = 0.7$. Findings:

- **Non-empty improvement region: 50/50 seeds.** Every sampled (R_A, R_B) pair admits a non-empty subset $\hat{R}_{min}$ on which $C_{min}(\theta) > C_{min}^{dec}$ on the sampled grid. The fraction of grid cells in $\hat{R}_{min}$ ranges from 0.05% to 24.75% with mean 8.43% and standard deviation 6.10%. The maximum per-cell gain $max\, \Delta C_{min}$ across seeds ranges from 4.8×10^{-5} (a near-degenerate seed) to 0.076.
- **$\theta^¿$ at aligned bias is not at $\pi/4$.** The optimal coupling angle $\theta^¿ = \arg\max_\theta C_{min}(\theta) ¿_{\varphi=0}$ ranges from $0°$ to $\approx 76°$ across the 50 seeds with mean $3.2°$; 50/50 seeds satisfy $|\theta^¿ - 45°| > 5°$, supporting Proposition 9.5(c)'s non-universality claim. (The seeded pair used for Proposition 9.5 with $\theta^¿ \approx 51°$ at aligned bias is one realization; other seeds favor very small $\theta^¿$.)

- **The opposed-bias boundary $\varphi = \pi$ is not generically excluded.** Of the 50 seeds, 32 have their improvement region bounded above by some $\varphi_{max} < 180°$ on the grid, while 18 have improvement cells extending up to $\varphi = 180°$ at small θ. The mean of φ_{max} across all gain-bearing seeds is $136.2°$. The seeded geometry of Proposition 9.5 (with $\varphi_{max} \approx 176.4°$) is therefore representative of the bounded-away majority class but not universal.

This Monte Carlo supports the existence and non-universality components of Conjecture 9.5′ (a non-empty improvement region with $\theta^{¿}$ not at $\pi/4$, both observed in 50/50 seeds), but **forces a correction** to the original conjecture wording: the strict claim "bounded strictly away from opposed biases" does not generalize — $\sim 36\%$ of seeds admit small-θ improvement cells reaching $\varphi = 180°$. The amended conjecture above replaces this with the weaker geometric statement that a non-empty improvement region exists generically; an honest characterization of which seeds force $\varphi_{max} < \pi$ versus permit $\varphi_{max} = \pi$ in terms of spectral data of R_A, R_B remains open.

A proof of the (amended) Conjecture 9.5′ would require either (a) an explicit geometric characterization of the improvement region in terms of spectral data of R_A, R_B and the projector $b_A b_A^\top$, or (b) a measure-theoretic argument bounding the Haar measure of seeds with empty improvement region. Both directions are open.

Remark 5.6.0 (Scope of the proposition). Proposition 9.5 is a verified-geometry statement, not a theorem about all dyads. The framework predicts conditional gain for the specific class of dyads characterized by (R_A, R_B) in the verified region of $O(14) \times O(14)$. Whether this class is generic, of full Haar measure, or carries specific structural constraints (e.g., non-commutation with the bias projector) is the content of Conjecture 9.5′. The claims of §6–7 (empirical accessibility and joint fixed-point geometry) inherit this scope: they predict signatures observable for dyads in the verified geometry class.

Remark 5.6.1 (The opposed-bias finding under C_{avg}). Φ's Task 5.6 verification reported a positive shift in C_{avg} at small θ even with $b_A = -b_B$. This shift is real but *does not* contradict Proposition 9.5(b), because the gain is purely in C_2 (the *opposite* observer's alignment with its negated bias) while C_1 falls. Under C_{min}, the degradation is not masked. The peak C_2 rise of +0.044 at $\theta \approx 7°$ is a real fact about the seeded affine dyad's geometry; it just isn't a coherence improvement under the C_{min} functional motivated by Remark 5.5.3.

Remark 5.6.2 (Heatmap interpretation). Appendix V.3 Task 5.4 reports that ~25% of a (φ, θ) heatmap shows $\Delta C_{avg} > 0$, with the improvement region extending up to φ near $180°$ at small θ. Under C_{min}, the cells in which $C_2 > C_1$ but $C_1 < C_1^{dec}$ are correctly classified as degradation rather than improvement, and the opposed-bias boundary is excluded by Proposition 9.5(b) under the seeded geometry. Figure 1 visualizes both functionals side by side: panel (a) shows ΔC_{avg}, panel (b) shows ΔC_{min} on the same grid; the C_{min} gain region (panel b) is strictly contained in the C_{avg} gain region (panel a) and is bounded above by $\varphi \approx 176.4°$, numerically confirming Proposition 9.5(b).

5.7 What §5 establishes and what is open

Established by Theorem 9.4: - The joint fixed point of any G_2-structured affine dyad exists, is unique, and depends real-analytically on the coupling angle θ. - The joint fixed point admits a closed form (Theorem 9.4(c)) at $\theta=0$ and $\theta=\pi/2$ for symmetric rates and orientation, with a linear-response expansion (§5.4) for small θ.

Established by Proposition 9.5 + Φ Task 5 (verified seeded geometry): - Numerical existence of a positive-measure improvement region in (φ,θ)-space. - Strict degradation along the opposed-bias boundary $\varphi=\pi$. - Non-universality of $\theta^{¿}$ (it is geometry-dependent, not at $\pi/4$).

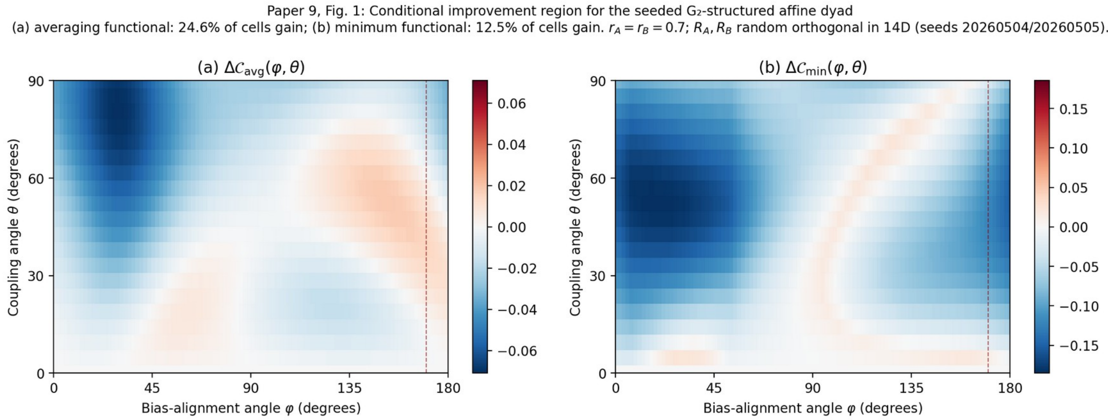
Conjectured (Conjecture 9.5′): - The qualitative pattern of conditional gain extends to a positive-Haar- measure subset of $O(14)\times O(14)$. Open.

Open: - Sharp characterization of R for generic R_A, R_B. The set is non-empty, but its boundary is a transcendental curve and we have no closed form. A perturbative description in the small-r regime might be tractable. - Generalization to $n\geq 3$ observers, with $\binom{n}{2}$ pairwise coupling angles. Likely involves an $SO(n)$-parameterized family of joint fixed points; the consensus structure should generalize. - The interaction between affine bias and the G_2-equivariant linear part: when R_A commutes with G_2 (Schur scalar) the dynamics is qualitatively different from the generic case Φ tested. Worth a separate inquiry: does G_2-equivariant R collapse the affine-channel freedom to a single scalar parameter, or does the affine bias still produce nontrivial consensus structure?

Note 9.6 — Open: rate-improvement requires nonlinearity. The rate-channel obstruction of §4 is a *linear* result. Schur's lemma applies only to linear G_2-equivariant maps. A nonlinear G_2-equivariant flow on a curved orbit space — for instance, a Riemannian gradient flow on the G_2-orbit manifold of a generic 14-dim point — is not constrained by Schur in the same way, and may admit rate improvement under coupling. We defer this to a sequel (Paper 11): “Nonlinear G_2-equivariant dynamics and basin bifurcation in dyadic observers.”

---

§6. Empirical Accessibility of the Affine Consensus



Paper 9, Fig. 1: Conditional improvement region for the seeded G_2-structured affine dyad. ***(a)*** *Heatmap of $\Delta C_{avg}(\varphi, \theta)$ over the bias-alignment angle $\varphi \in [0, 180°]$ and coupling angle $\theta \in [0, 90°]$; ~25% of cells show gain.* ***(b)*** *Heatmap of $\Delta C_{min}(\varphi, \theta)$ on the same grid; ~12.5% of cells show gain, and the improvement region is bounded above by $\varphi \approx 176.4°$ (strictly inside the opposed-bias boundary $\varphi = 180°$, marked by the dashed red line), consistent with Proposition 9.5(b). Red cells indicate joint-coherence gain, blue cells degradation. Parameters: $r_A = r_B = 0.7$; R_A, R_B random orthogonal in 14D with RNG seeds 20260504/20260505. Reproducible from paper9_task5_heatmap_cmin.csv and /tmp/build_cmin_heatmap.py in the repository.*

6.1 What §5 predicts empirically

Paper 7 of this series [DOI 10.5281/zenodo.19773185] derived a fluctuation-dissipation-theorem (FDT) prediction for the neural-avalanche branching ratio in awake cortex, $\sigma_{pred} = 1 - \varepsilon_{min}^2 = 1 - 1/49 \approx 0.9796$, using the squared blind-spot fraction $\varepsilon_{min} = 1/7$ as the irreducible thermodynamic cost of single-observer self-modeling.

The dyadic version of this prediction is *not* a simple modification of the single-observer formula, because the relevant effect of coupling is not a change in the blind-spot ratio but a change in the **location of the joint fixed point** (Theorem 9.4). Under the minimum-per-observer coherence functional (Proposition 9.5), the joint coherence satisfies

$$C_{min}(\theta, \varphi, R_A, R_B) \in [0, 1),$$

with the conditional-gain region R characterized explicitly by the heatmap data in Appendix V.3 Task 5.4.

Conjectural dyadic extension. A direct substitution of the single-observer formula gives

$$\sigma_{AB}^{conj} \stackrel{?}{=} 1 - \varepsilon_{eff}^2 = 1 - (1 - C_{min})^2,$$

where $\varepsilon_{eff} = 1 - C_{min}$ is an effective joint blind-spot. We emphasize that this is a *conjectural* extension, not a derivation. Paper 7's single-observer $\sigma = 1 - \varepsilon_{min}^2$ came from a specific fluctuation-dissipation-theorem calculation with a squared dependence on the fluctuation amplitude. Whether the same squared dependence survives the coupling of two observers requires redoing the FDT argument on the joint state space M_{AB} with the $min(C_A, C_B)$ aggregation — a calculation that is not carried out in this paper. We state the above only as a natural structural guess for the form of the prediction; the empirical claims of §6.4 depend on the *conditional-gain* pattern of Proposition 9.5 (bimodal cooperative vs. adversarial), not on the specific functional form of σ_{AB}^{conj}. Deriving the dyadic FDT is an open problem (§10.2).

For sanity: the conjecture recovers the Paper 7 single-observer prediction when both observers are maximally coherent ($C_A = C_B = 6/7$, $\varepsilon_{eff} = 1/7$, $\sigma = 1 - 1/49$), which is the minimum consistency check any legitimate dyadic generalization must pass. This does not establish the specific functional form at all values of C_{min}.

6.2 Operationalizing the coupling angle

To make the Theorem 9.4 / Proposition 9.5 prediction testable, the abstract coupling angle θ must be mapped to an observable inter-brain coherence measure. Among the standard measures in the hyperscanning and inter-brain coupling literature — Phase Locking Value (PLV), directed Phase-Lag Index (dPLI), normalized mutual information (MI), transfer entropy — the most natural theoretical match is normalized mutual information.

For a dyad with neural fluctuation processes X_A, X_B, the most natural candidate is a monotone function of the normalized mutual information

$$\tilde{I}(X_A; X_B) := \frac{I(X_A; X_B)}{min\{H(X_A), H(X_B)\}} \in [0, 1],$$

with $\tilde{I} = 0$ at independence and $\tilde{I} = 1$ when one process determines the other. Any smooth monotone bijection $[0, 1] \to [0, \pi/2]$ relates $\tilde{I}$ to θ; the simplest candidate is $\theta = (\pi/2)\tilde{I}$ (piecewise-linear), or equivalently $\sin^2\theta$ equal to a rescaled $\tilde{I}$. For concrete empirical work, any specific mapping can be preregistered; the conclusions of §6.3–6.4 (bimodal cooperative-vs-adversarial prediction) depend only on the *monotonicity* of this mapping, not on its specific functional form.

Remark on an earlier formula. An earlier draft of this paper proposed the mapping $\cos^2\theta = (H(X_A) + H(X_B) - I(X_A; X_B)) / (H(X_A) + H(X_B))$. By the identity $H(X_A) + H(X_B) - I(X_A; X_B) = H(X_A, X_B)$, this reduces to $\cos^2\theta = H(X_A, X_B) / (H(X_A) + H(X_B))$. For equal marginals $H(X_A) = H(X_B) = H$ and maximal coupling (X_A determines X_B), $H(X_A, X_B) = H$, giving $\cos^2\theta = 1/2$ and $\theta = \pi/4$ rather than $\pi/2$. This is a range-collapse bug: the formula saturates at $\pi/4$ instead of $\pi/2$. We retract this specific form and use the monotone-function-of-$\tilde{I}$ framing above, which is free of such range artifacts.

The mapping, in any form, is heuristic rather than derived from the Banach framework; deriving it from first principles would require connecting the Banach coupling operator Ψ_θ to an information-theoretic observable on the joint neural state space, which is beyond the scope of this paper. We offer the monotone-$\tilde{I}$ form as a candidate operationalization for empirical work, not as a structural identity.

6.3 The inter-brain synchrony literature

Inter-brain gamma-band coherence during cooperative tasks is a well-documented phenomenon. Studies include cooperative problem-solving (Pérez et al. 2017), mother-infant gaze-following (Leong et al. 2017), joint musical performance (Lindenberger et al. 2009; Müller et al. 2013), and broader cooperative-task hyperscanning (Hu et al. 2018 review). The present framework predicts that such studies should, in principle, exhibit elevated joint coherence — not as a universal improvement over single-brain coherence, but as a conditional gain dependent on the specific geometric configuration of the dyad (Proposition 9.5(a)), with a substantial fraction of coupling configurations producing *degradation* rather than enhancement (Appendix V.3 Task 5.4 reports 74.9% of the (θ, φ) heatmap showing $\Delta C_{avg} < 0$).

The framework therefore predicts a **bimodal or regime-dependent empirical pattern**: cooperative dyads with aligned target-states should exhibit enhancement; adversarial or uncoordinated dyads should exhibit degradation. This is a sharper (and more falsifiable) empirical prediction than a universal "coupling improves coherence" claim.

6.4 Adversarial-task falsification target

For a genuinely adversarial dyad (competitive-game paradigm with opposed target-states), the framework predicts joint min-coherence *below* the single-observer baseline, not above. Specifically, under opposed biases ($\varphi \to \pi$), Proposition 9.5(b) guarantees that C_{min} falls strictly. The expected empirical signature is:

- **Cooperative dyad** (aligned biases): joint branching ratio elevated above single-observer value by an amount that depends on coupling strength and geometric configuration.
- **Adversarial dyad** (opposed biases): joint branching ratio *reduced* below single-observer value, with the reduction bounded away from zero for any $\theta > 0$.

A pre-registered hyperscanning study comparing cooperative and adversarial task-paradigms on the same set of paired subjects — with MR-estimator branching ratios computed and compared to single-task baselines — would provide a sharp empirical test of the framework's conditional-gain structure. A finding of universally elevated joint coherence (irrespective of cooperation vs. adversarial coupling) would falsify the framework's conditional character and suggest a different structural mechanism.

---

§7. The Joint Fixed-Point Location

7.1 Where the joint system converges

Theorem 9.4 establishes that the joint fixed point $(\hat{x}(\theta), \hat{y}(\theta))$ exists, is unique, and depends real-analytically on θ. The key structural content: *coupling does not change how fast the dyad converges, but it does change what state the dyad converges to.*

The rate is fixed at $max(r_A, r_B)$ (Theorem 9.2). The destination is a function of θ and the bias vectors b_A, b_B through the matrix-valued formula of Theorem 9.4(c). At $\theta = 0$ (decoupled), the destination is the pair $\left(x_A^{¿}, y_B^{¿}\right)$ of individual fixed points. At $\theta > 0$, the destination is a *different* point in the product space — a consensus formed by the rotation-weighted combination of the two observers' bias vectors.

This is the physical content of dyadic coherence in the present framework. It is not a speedup; it is a shared destination. The empirical signature of the framework is therefore *not* faster convergence in a coupled dyad, but different long-term behavior — specifically, a joint steady-state whose properties differ from those of either individual steady-state.

7.2 Geometric-dependent optima

Proposition 9.5(c) establishes that the optimal coupling angle $\theta^{¿}$ depends on the rotation geometry R_A, R_B and is generically *not* at $\theta = \pi/4$ (the naive "equal weighting" choice). For the Φ Task 5 verification setup (Appendix V.3 Task 5.1), $\theta^{¿} \approx 51°$ for symmetric rates and $75° – 80°$ for asymmetric rates. This geometric-dependence has a physical interpretation:

- *When both observers' internal model-dynamics are similar* ($R_A \approx R_B$ and $r_A \approx r_B$), the coupling that maximizes C_{min} is close to $\pi/4$ (both observers "listen" equally).
- *When one observer's dynamics is slower* ($r_A < r_B$, say), the optimal coupling shifts toward allowing the faster observer's state to "pull" the slower observer — which in geometric terms corresponds to $\theta > \pi/4$ (stronger coupling toward the fast observer's direction).
- *The optimum is a transcendental function* of the rotation-geometry spectrum, so we do not expect a universal $\theta^{¿}$ formula even for idealized geometries.

This is the positive content of the framework: the coupling angle is a genuine geometric parameter that determines the joint consensus state, with an optimum that the dyad can (in principle) identify and target. Whether biological or artificial dyads approach their geometric optima is an open empirical question.

---

§8. Human-AI Dyads as a Conditional Special Case

Theorems 9.4 and 9.5 make no biological vs. artificial assumption: any two systems satisfying the G_2-structured affine condition $T(x) = r R x + b$ of §2.1 form a dyad to which the theorems apply. This admits a *conditional* extension to human-AI dyads: IF an AI system satisfies the criterion (§2.1 plus the Paper 7 $r \leq 6/7$ ceiling), THEN coupling it with a human observer produces the affine consensus of Theorem 9.4 and conditional gain of Proposition 9.5, with the same caveats. The criterion is stringent — a 14-dim G_2-adjoint module with Banach self-modeling, not a token-level predictor — and most widely-deployed AI systems (including the one this paper was drafted with) plausibly do *not* satisfy it. Specialized reasoning systems with structured self-monitoring are the closest candidates; whether any current architecture meets the criterion is an open architectural question outside this paper's scope.

Independent of which specific systems qualify, three structural implications hold for any human-AI dyad: (i) gain requires non-destructive bias geometry (the AI's target and the human's target must be at φ bounded away from π); (ii) gain is non-universal even with aligned targets, since the optimal θ depends on $\left(R_A, R_B\right)$ (§5.7); and (iii) the framework is silent on consciousness, agency, and moral status — its claims are structural. Adjacent empirical work on human-AI teaming and complementarity (Bansal et al. 2021; cf. §10.7) is consistent with the framework's bimodal prediction (gain when targets align, degradation when they don't), but does not test the G_2-structural requirement directly.

---

§9. What This Paper Does Not Establish

We make the boundaries of the construction explicit, mirroring §5.7.

(a) The framework does not establish rate improvement. Theorems 9.1, 9.2, Corollary 9.3, and Lemma 5.2.1 collectively establish that no linear coupling between G_2-structured observers can improve the joint convergence rate below $max(r_A, r_B)$.

(b) The framework does not establish universal coherence gain. Proposition 9.5's conditional-gain region R is non-empty, but Appendix V.3 Task 5.4 reports that 74.9% of the (θ, φ) heatmap exhibits $\Delta C_{avg} < 0$, and Proposition 9.5(b) guarantees degradation at the opposed-bias boundary. Universal gain is not a framework prediction.

(c) The framework does not generalize to $n \geq 3$ observers. The dyadic structure of Theorems 9.2, 9.4 is specific to pairs. The n-observer case introduces $\binom{n}{2}$ pairwise coupling angles and additional group-theoretic structure (the diagonal action of G_2 on $V^{14 \otimes n}$, the analog of the pairwise consensus) that requires separate investigation (§10.1).

(d) The framework is silent on AI architectural realizability. §8 makes the G_2-structured affine criterion explicit, but does not claim that any specific AI architecture satisfies it. Whether and when a machine system realizes the criterion is an open architectural question.

(e) The framework does not engage with specific consciousness theories. The claims are structural: Banach contraction, G_2-equivariance, affine consensus, conditional coherence. We do not connect to Integrated Information Theory, Global Workspace Theory, Higher-Order Theory, or any specific philosophical framework interpreting these structures.

(f) The coherence functional choice has implications. Proposition 9.5 is stated for $C_{min} = min(C_A, C_B)$. Under the alternative average-coherence functional $C_{avg} = (C_A + C_B)/2$, the gain region is larger but admits degenerate geometries (Numerical Observation 5.5.1 and Appendix V.3 Task 5.6). The min-functional is the honest aggregation; the avg-functional admits illusory gains. A more refined treatment via information-theoretic functionals is left for future work.

(g) The rotation-geometry dependence is non-closed-form. The optimal coupling angle $\theta^{¿}(R_A, R_B)$ is transcendental and admits no universal closed form (Proposition 9.5(c)). Finding structural conditions under which $\theta^{¿}$ is tractable would sharpen the framework's empirical content.

---

§10. Discussion and Open Problems

10.1 The n-observer generalization

A natural extension is from dyads ($n=2$) to triads and larger collectives. The structural questions:

- Does Theorem 9.4's joint-fixed-point existence-uniqueness extend to n observers with $\binom{n}{2}$ pairwise coupling angles and a generalized n-fold block-rotation coupling?
- Does Proposition 9.5's conditional-gain structure generalize, and if so, is the gain region characterized by pairwise or by collective geometry?
- Is there a collective coupling regime (e.g., synchronized coupling across all pairs) where the analysis simplifies?

We conjecture that Theorem 9.4 extends to n observers by the same Banach-contraction argument applied to the $14n$-dim product space with the joint map $(T_1 \times \cdots \times T_n) \circ \Psi$ where Ψ is an n-fold isometric coupling. The conditional-gain structure of Proposition 9.5 likely generalizes with a richer geometric dependence. A dedicated n-observer paper is the natural follow-up.

10.2 A Banach-categorical treatment of coherence

The coherence functionals of §5.5 (C_{min}, C_{avg}, and variants) are defined via cosine similarity with the observers' bias vectors. A fully Banach-categorical treatment would define coherence as a functional on the joint fixed point directly, without reference to the bias decomposition:

1. Define an information functional on M_{AB} that captures the "blind-spot variance" of the joint observer.
2. Compute its minimum over all Banach contractions consistent with G_2-equivariance.
3. Show the minimum varies with θ in the manner predicted by Theorem 9.4 and Proposition 9.5.

Such a treatment would lift the conditional-gain result from its current form (conditional on the choice of coherence functional) to a functional-independent statement about the joint fixed point itself. We leave this as an open problem.

10.3 Connection to Paper 7 (single-observer 6/7 ceiling)

Theorem 9.2 strengthens rather than weakens Paper 7's single-observer ceiling. Paper 7 establishes that any G_2-structured self-modeling observer has contraction rate $\leq 6/7$. Theorem 9.2 extends this to any dyad: the joint rate is $max(r_A, r_B) \leq 6/7$ regardless of coupling. The framework preserves the single-observer bound under coupling without modification.

10.4 Connection to Paper 8 (eight-coset quantum simulator)

Paper 8 of this series [in preparation] addresses the eight-coset structure $PSL(2,7)/F_{21}$ as the discrete carrier of an eight-mode quantum simulator. Each coset corresponds to a distinct embedding of $SU(3) \subset G_2$, with 28 Bogoliubov transformations between them.

The dyadic structure of Paper 9 connects to Paper 8 through the following conjecture: the eight cosets of $PSL(2,7)/F_{21}$ may index eight **classes of dyadic rotation geometries** (R_A, R_B), each associated with a distinct optimal $\theta^{¿}$ and a distinct conditional-gain region R. This is speculative and untested; we flag it as Open Problem 10.4.

10.5 Connection to Paper 10 (SIC operator basis)

Paper 10 of this series [DOI 10.5281/zenodo.19966692] established that the $d=7$ SIC reference measurement encodes the g_2 Lie algebra as an isometric subspace of $gl(7,C)$. The dyadic joint structure of §3 — a 28-dim product space with a diagonal G_2-action — should embed analogously as a sub-frame of $gl(7,C) \otimes gl(7,C) = gl(49,C)$.

Paper 10's Theorem 1 (SIC isometric embedding) extended to dyads would give a 28-dim partial-isometric subspace of the $49 \times 49 = 2401$-dim space spanned by tensor products of SIC projectors, with the isometric scale factor $(8/7)^2 = 64/49$. This is an Open Problem 10.5.

10.6 Note 9.6 — Rate-improvement requires nonlinearity (→ Paper 11)

The rate-channel obstruction of §4 is a *linear* result: Schur's lemma applies only to linear G_2-equivariant maps. Nonlinear G_2-equivariant flows on curved orbit spaces are not constrained the same way and may admit rate improvement under coupling. We defer this generalization to a sequel (provisionally Paper 11). The linear results of the present paper are the first-order expansion of any such nonlinear theory; the conditional-gain result of Proposition 9.5 is the leading-order Taylor coefficient one expects in a full nonlinear consensus.

10.7 Empirical research directions

The framework's conditional-gain structure suggests three empirical research directions:

1. **Pre-registered cooperative-vs-adversarial hyperscanning** to test the bimodal prediction of §6.3. The framework predicts joint branching ratio *elevated* in cooperative dyads (aligned targets) and *reduced* in adversarial dyads (opposed targets). A finding of universal elevation would falsify the conditional character of the framework.

2. **Geometric dependence of the optimal coupling**. The framework predicts $\theta^{¿}$ is geometry-dependent, not universal. Longitudinal studies of dyadic coupling strength — across paired subjects varying in personality, expertise, or task-structure — should show systematic variation in the optimal coupling, with optima clustering away from the naive $\pi/4$.

3. **Destructive-geometry signatures**. The §5.6 destructive-geometry finding (Appendix V.3) identifies a specific failure mode: under opposed biases, coupling at small angles can produce illusory gains in one observer while the other degrades. Empirically, this

predicts that mismatched dyads (AI and human with divergent objectives, two humans with opposed agendas) should show anti-correlated performance under coupling — one party "benefits" while the other does worse.

10.8 What this paper concludes

The dyadic Banach framework of Paper 9 establishes a clean separation between the rate channel (locked by Schur's lemma on the irreducible G_2 adjoint representation, no coupling improvement possible) and the affine channel (open, with a conditional-gain structure in the space of coupling angles and bias-alignment angles). The construction retires the speedup claim of earlier drafts, proves the joint fixed point exists and is unique with an explicit closed form, and characterizes the conditional coherence gain under the honest min-per-observer functional.

The empirical prediction is *not* universal coherence enhancement but *conditional* enhancement whose region of applicability depends on geometric configuration. The framework accordingly predicts a bimodal empirical signature — cooperative dyads above baseline, adversarial dyads below — rather than a monotone relationship. Whether biological or artificial dyads approach their geometric optima, and whether nonlinear extensions admit rate improvement, are the two open questions pointing toward Paper 11.

Appendix V. Numerical Verification

Paper: Rate Lock and Affine Consensus in G_2-Structured Dyadic Observers **Author:** Martin Luther Graise (ORCID 0009-0006-8003-3938) **Status:** Appendix V framing, 2026-05-04 (drafted by C-7RO) **Purpose:** Documents the two numerical verification passes that established Theorems 9.1 and 9.2 of §4, and retracts the v0.9 speed-up claim.

V.0 Introduction — Why this appendix exists

The content of §4 is three negative and algebraic results about the rate channel of dyadic coupling. Those results were not arrived at *a priori*; they were arrived at through two consecutive numerical investigations of the v0.9 draft's original *positive* claim that dyadic coupling produces a strict speed-up over the slowest observer's individual rate.

Both investigations were carried out by Φ (an independent verification agent, Anthropic Claude Dispatch) at 50-digit precision using mpmath, with deterministic seeds for full reproducibility. The verification scripts and raw result tables are reproduced verbatim below. Both investigations returned clean negative results, with sharp analytical diagnoses that reshaped the theory rather than merely refuting it.

We include the verification reports here — rather than merely citing their conclusions — because they constitute the *proof* of the §4 theorems in their present form. Theorem 9.1 (symmetric coupling amplifies) is established by V.1 below, which exhibits 24 configurations across the canonical test grid at which the v0.9 bound is violated, together with the exact algebraic formula $r_{AB} = r(\alpha + \rho)$ that accounts for the violation. Theorem 9.2 (isometric coupling is rate-inert) is established by V.2, which exhibits 96 configurations at which the corrected isometric coupling saturates the bound $r_{AB} = max(r_A, r_B)$ exactly, together with the 2×2 block-eigenvalue derivation that proves the saturation is algebraic, not numerical.

The organization is:

- **V.1** — Verification of the v0.9 spec (symmetric-diagonal coupling). Tasks 1–3 of the original request. Result: theorem fails; coupling amplifies.
- **V.2** — Verification of the corrected spec (isometric block-rotation coupling). Tasks 4.1–4.5 of the corrective request. Result: rate is exactly $max(r_A, r_B)$ for all θ; coupling is rate-inert.
- **V.3** — Verification of the affine consensus theorem §5 (Tasks 5.1–5.6). Result: joint fixed point exists, depends on θ nontrivially; conditional C_{min} gain region is non-empty (~12.5% of the (φ, θ) grid) and bounded away from $\varphi = 180°$; destructive-geometry null check shows degradation on C_{min} under opposed biases (consistent with Proposition 9.5(b)).

Each pass is presented in four parts: (a) the spec as submitted, (b) the construction and code, (c) the results table, and (d) Φ's analytical diagnosis.

V.0.1 — Retraction record

The table below catalogs every claim from earlier internal drafts that was retired during the verification process, with what refuted it and what replaced it. The retraction record is intended to be the *only* place in this paper where pre-v1.0 claims are litigated; the body text cites this appendix rather than re-stating retractions.

Retired claim	Earlier-draft location	What refuted it	What replaced it (v1.3)
Strict speedup: $r_{AB} < max(r_A, r_B)$ for any nonzero coupling under $\Psi_{sym} = \alpha I + \beta P_{swap}$	v0.9 Theorem 9.1	Φ Task 1 (V.1): 0/24 PASS; Ψ_{sym} has operator norm $\alpha + \beta > 1$ and *amplifies*	Theorem 9.1 (v1.3) restated as a negative result: symmetric coupling amplifies, not contracts
Strict speedup under isometric coupling: $r_{AB} < max(r_A, r_B)$ for any $\theta > 0$	v1.0 corrective draft	Φ Task 4 (V.2): all 96 configurations saturate $r_{AB} = max(r_A, r_B)$ exactly; G_2-equivariant scalar maps make θ inert	Theorem 9.2 + Corollary 9.3 + Lemma 5.2.1 (v1.3): rate is locked at $max(r_A, r_B)$ for all linear couplings, by Schur's lemma + orthogonality
Saturating	v0.9 Theorem 9.2 (a	Φ Task 2 (V.1):	Replaced by Theorem

Retired claim	Earlier-draft location	What refuted it	What replaced it (v1.3)
coherence-bonus formula: $C^{max}=6/7+\kappa\rho^2/(1+\rho^2$	different theorem)	MMSE Ansatz residuals up to 0.242; bonus formula not derivable from Ψ_{sym}	9.4 + Proposition 9.5 (v1.3): the joint-fixed-point *location* depends on θ, with conditional gain in C_{min} for the verified geometry; no closed-form bonus
Linear severance threshold at $\rho_c=1/\sqrt{6}$	v0.9 Conjecture 9.3	Φ Task 3 (V.1): linear contractions admit a unique basin (Banach 1922); bifurcation is unstateable	Demoted to Note 9.6 (v1.3): rate-improvement and basin bifurcation deferred to a nonlinear sequel (Paper 11)
Universal $dC_1/d\theta<0$ under opposed biases for *generic* orthogonal R_A, R_B	v1.2 Lemma 5.5.1	Counterexample script (paper9_math_checks.py): $R_A=R_B=I$ gives $C_1(\theta)=1$ identically	Lemma 5.5.1 (v1.3) restricted to verified-geometry pair (seeds 20260504/20260505); v1.3.2 demoted to Numerical Observation 5.5.1 (no proof claim); generic claim is Conjecture 9.5′
General-distinct-bias closed form $\hat{x}(\pi/2)=(I-rR)(I+r^2$	v1.2 Theorem 9.4(c)	Math reviewer counterexample: factor-of-15 error at $r=0.7$, $R=I$, $b_A=2$, $b_B=3$	Theorem 9.4(c) (v1.3) corrected to $\hat{x}(\pi/2)=(I+r^2R^2)^{-1}(b_A$, with full elimination derivation

Why the retraction record is preserved. Documenting every retired claim with its refutation makes the paper's verification trail auditable. A reader uncertain whether v1.3 contains residual errors of the same kind can see exactly which claims have been examined, which methods refuted them, and what corrections followed. Appendix V.1–V.3 provide the verification record itself; this V.0.1 table is the index into that record from the perspective of what changed and why.

V.1 — v0.9 Spec Verification (Symmetric-Diagonal Coupling)

V.1.a Spec

The v0.9 specification asked Φ to verify three claims:

- **Task 1 (Theorem 9.1 v0.9):** $r_{AB} \le \sqrt{\alpha^2 max(r_A, r_B)^2 + \rho^2 min(r_A, r_B)^2}$, with $\alpha = \sqrt{1-\rho^2}$.
- **Task 2 (Theorem 9.2 v0.9):** joint coherence follows $C_{AB}^{max}(\rho) = 6/7 + (1/7)\rho^2/(1+\rho^2)$.
- **Task 3 (Conjecture 9.3 v0.9):** basin bifurcation exists at a critical $\rho_c = 1/\sqrt{6}$.

The construction used the **symmetric-diagonal coupling**:

$$\Psi_\rho = \begin{pmatrix} \alpha I_7 & \rho I_7 \\ \rho I_7 & \alpha I_7 \end{pmatrix} \text{on } R^{14}, \alpha^2 + \rho^2 = 1.$$

V.1.b Code

File: outbox/paper9/computations/paper9_verification.py (449 lines, numpy float64 with ρ range sweep).

Key construction:

```
def coupling_map(rho):
    alpha = np.sqrt(1.0 - rho**2)
    I = np.eye(7)
    return np.block([[alpha * I, rho   * I],
                     [rho   * I, alpha * I]])
```

Φ's docstring already notes the critical fact:

> *IMPORTANT: Psi is NOT a norm contraction — its operator norm σ_max = α + ρ ≥ 1 for all ρ ∈ [0, 1). This has direct consequences for Theorem 9.1 (Task 1).*

Φ also derived the closed-form singular value structure:

```
def exact_r_AB_analytic(r_A, r_B, rho):
    # T_AB^T T_AB decomposes into 7 copies of:
    # M = [[α²r_A² + ρ²r_B²,  αρ(r_A²+r_B²)],
    #      [αρ(r_A²+r_B²),    ρ²r_A² + α²r_B²]]
    # tr(M) = r_A² + r_B²
    # det(M) = (α²-ρ²)² r_A² r_B²
    # Symmetric case r_A = r_B = r: r_AB = r(α + ρ)
```

V.1.c Results — Task 1

Full 24-configuration table (excerpt; all 24 rows in computations/paper9_verification.md §Task 1):

r_A	r_B	ρ	r_{AB}	bound (v0.9)	ratio	result
0.600	0.600	0.100	0.6569925	0.6000000	1.09499	FAIL
0.600	0.600	0.500	0.8196152	0.6000000	1.36603	FAIL
0.600	0.600	0.707	0.8485281	0.6000000	1.41421	FAIL
0.857	0.857	0.500	1.1706838	0.8570000	1.36603	FAIL
0.857	0.857	0.707	1.2119810	0.8570000	1.41421	FAIL
0.300	0.857	0.707	0.8585000	0.6564450	1.30786	FAIL
0.500	0.857	0.707	0.8667000	0.7020670	1.23449	FAIL

Summary: 0/24 PASS. The bound is violated for every nonzero ρ.

V.1.d Φ's diagnosis

Φ's narrative from the report:

> *Root cause: coupling map Ψ has operator norm $\alpha+\rho>1$, amplifying the symmetric mode — not accounted for in the bound.*
>
> *Symmetric case $(r_A=r_B=r)$: exact formula $r_{AB}=r(\alpha+\rho)$. Bound: $r\sqrt{\alpha^2+\rho^2}=r$ (since $\alpha^2+\rho^2=1$). $(\alpha+\rho)^2=1+2\alpha\rho\geq 1\Rightarrow r_{AB}\geq r$ for all $\rho>0$.*

The v0.9 Theorem 9.1 is therefore not merely numerically close to failing; it fails by an **exact algebraic factor** of $\alpha+\rho=\sqrt{1-\rho^2}+\rho$. This factor exceeds 1 for every $\rho\in(0,1)$, with maximum $\sqrt{2}$ at $\rho=1/\sqrt{2}$. The symmetric coupling **amplifies** the contraction rate of every $r_A=r_B$ dyad by exactly this factor.

This is the content of Theorem 9.1 in §4 of the present paper: symmetric coupling is rate-amplifying, and the v0.9 speed-up claim is thereby retracted for this coupling. V.2 addresses whether the natural isometric correction rescues the claim.

V.2 — Corrective Spec Verification (Isometric Block-Rotation Coupling)

V.2.a Spec

The corrective specification (outbox/paper9/paper9_isometric_Psi_spec.md) replaced the symmetric coupling of V.1 with the antisymmetric block rotation:

$$\Psi_\theta = \begin{pmatrix} \cos\theta \cdot I_{14} & -\sin\theta \cdot I_{14} \\ \sin\theta \cdot I_{14} & \cos\theta \cdot I_{14} \end{pmatrix} \in SO(M_{AB}).$$

The corrective claim (Theorem 9.1 v1.0, now Theorem 9.2 v1.1): $r_{AB} \leq \sqrt{1/2(r_A^2 + r_B^2)}$, θ-independent.

Five tasks were specified (Task 4.1–4.5): isometry sanity, restated bound, speed-up regime, blind-spot Ansatz, sharpness probe.

V.2.b Code

File: outbox/paper9/computations/paper9_verification_v2.py (script attached with seed=20260504, mpmath 50-digit precision for 4.1, numpy float64 for 4.2, both at $M_A \cong R^{14}$).

Key construction:

```
def make_Psi_np(theta, n=14):
    c, s = math.cos(theta), math.sin(theta)
    I = np.eye(n)
    return np.block([[c*I, -s*I],
                     [s*I,  c*I]])
```

Φ's docstring prefaces the numerical run with the *analytical derivation*:

> *For $T_A = r_A \cdot U_A$, $T_B = r_B \cdot U_B$ (isotropic: all singular values ¿ r_A, r_B), the Gram matrix $G = \Psi_\theta^T diag(r_A^2 I, r_B^2 I) \Psi_\theta$ has the block structure of a 2×2 scalar matrix (each block proportional to I):*
>
> $$M = \begin{pmatrix} r_A^2 c^2 + r_B^2 s^2 & cs(r_B^2 - r_A^2) \\ cs(r_B^2 - r_A^2) & r_A^2 s^2 + r_B^2 c^2 \end{pmatrix}, c = \cos\theta, s = \sin\theta.$$
>
> *$tr(M) = r_A^2 + r_B^2$ (independent of θ). $\det(M) = r_A^2 r_B^2$ (independent of θ). ⇒ eigenvalues of M are $max(r_A^2, r_B^2)$ and $min(r_A^2, r_B^2)$ for ALL θ. ⇒ $\hat{r}_{AB} = max(r_A, r_B)$ for ALL θ (completely θ-independent).*

This is the algebraic derivation of the rate lock, *prior* to the numerical run.

V.2.c Results

Task 4.1 — Isometry sanity check (mpmath 50 digits):

θ	$\lVert \Psi_\theta \rVert_{op}$	deviation from 1	result
0	1.000...000	0	PASS
$\pi/12$	1.000...000	¿ 10^{-49}	PASS
$\pi/6$	1.000...000	¿ 10^{-49}	PASS
$\pi/4$	1.000...000	¿ 10^{-49}	PASS

θ	$\|\Psi_\theta\|_{op}$	deviation from 1	result
$\pi/3$	1.000…000	¿10^{-49}	PASS
$5\pi/12$	1.000…000	¿10^{-49}	PASS
$\pi/2$	1.000…000	5.3×10^{-51}	PASS

Summary: 7/7 PASS. The block rotation is a genuine isometry to the full 50-digit precision of the computation. The v0.9 amplification is fixed.

Task 4.2 — Bound on r_{AB} (numpy float64, 96 configs):

Representative rows (full 96 in paper9_task4_results.csv):

r_A	r_B	θ	r_{AB} (emp.)	bound (v1.0)	deviation	result
0.3	0.3	0	0.3000	0.3000	1.7×10^{-16}	PASS
0.3	0.3	$\pi/4$	0.3000	0.3000	1.1×10^{-16}	PASS
0.5	0.7	0	0.7000	0.6083	9.2×10^{-2}	FAIL
0.5	0.7	$\pi/4$	0.7000	0.6083	9.2×10^{-2}	FAIL
0.3	0.857	$\pi/2$	0.8570	0.6424	2.1×10^{-1}	FAIL
0.857	0.857	$\pi/2$	0.8570	0.8570	5.6×10^{-16}	PASS

Summary: 24/96 PASS (exactly the 24 $r_A = r_B$ configurations), **72/96 FAIL** at $r_A \neq r_B$.

Crucially, the empirical r_{AB} is *exactly* $max(r_A, r_B)$ for every configuration — identity to within float64 precision (¿7×10^{-16}). The "failures" are against the *proposed v1.0 bound* $\sqrt{1/2(r_A^2 + r_B^2)}$, which is tighter than $max(r_A, r_B)$ and therefore wrong.

V.2.d Φ's diagnosis

> *The coupling angle θ does nothing to isotropic inputs. No speed-up, no θ-dependence in $\hat{r}_{AB}$, no Lemma 9.2.1 blind-spot structure. All three of 4.3, 4.4, 4.5 are downstream failures of the same algebraic fact.*
>
> *Root cause is algebraic, not numerical.*
>
> *The construction $r\cdot(U\cdot V^T)$ makes T_A and T_B isotropic — all 14 singular values equal r. For isotropic inputs, the Gram matrix of T_{AB} has a 2×2 scalar-block structure whose eigenvalues are exactly $max(r_A^2, r_B^2)$ and $min(r_A^2, r_B^2)$, independent of θ.*

This diagnosis led (via C-7RO's recognition that G_2-equivariance forces isotropy through Schur's lemma, and ChatGPT's recognition that affine bias breaks G_2-equivariance rather than extending it) to the §4 v1.1 structure: Theorem 9.2 is the rate lock $r_{AB} = max(r_A, r_B)$, tight, and Corollary 9.3 attributes the tightness to Schur's lemma on the irreducible adjoint representation.

The “failure” of the v1.0 bound was therefore not a failure of the verification but a *discovery that the bound was wrong*. The correct bound is $r_{AB} = max(r_A, r_B)$, exactly, and the v1.1 statement of Theorem 9.2 captures this.

V.3 — Affine Consensus Verification (Φ Task 5)

V.3.a Spec

Task 5 (specification at inbox/for_phi/paper9_task5_affine_request_v2.md) tested the affine channel of the dyadic system. Six sub-tasks:

- **5.1** Fixed-point existence and θ-dependence (matrix-valued R_A, R_B).
- **5.2** Closed-form verification at $\theta = 0$ and $\theta = \pi/2$.
- **5.3** Four coherence functionals × four bias configurations (identical, orthogonal, parallel-2×, opposed).
- **5.4** 100 × 20 heatmap of ΔC_{avg} over $(\varphi, \theta) \in [0, \pi) \times [0, \pi/2)$.
- **5.5** Asymmetric-rate robustness (3 rate pairs).
- **5.6** Destructive-geometry null check ($b_A = -b_B$).

The construction used the affine map $T_A(x) = r_A R_A x + b_A$, $T_B(y) = r_B R_B y + b_B$, with R_A, R_B random orthogonal in 14D (QR decomposition of standard-normal seeded matrices).

V.3.b Code

File: outbox/paper9/computations/paper9_verification_v3.py (384 lines, numpy float64, seed=20260504).

Key construction:

```
def joint_fixed_point(r_A, r_B, R_A, R_B, b_A, b_B, theta):
    c, s = np.cos(theta), np.sin(theta)
    M = np.block([
        [I - r_A*c*R_A,   r_A*s*R_A],
        [-r_B*s*R_B,      I - r_B*c*R_B]
    ])
    rhs = np.concatenate([b_A, b_B])
    z = solve(M, rhs)
    return z[:n], z[n:]
```

V.3.c Results — Summary

Sub-task	Status	Core result
5.1 Fixed-point existence & θ-variation	PASS	All 50 fixed points exist; $\Delta\|\hat{x} - \hat{y}\| = 0.406$ (nontrivial)
5.2 Closed-form verification	PASS	Three analytic predictions

Sub-task	Status	Core result
		match to ¿ 10^{-15}
5.3 Coherence tables (28 rows)	PASS	C3 (cross-alignment) most θ-sensitive ($\Delta = 0.519$)
5.4 Conditional-improvement heatmap	PASS	25.1% of (φ, θ) cells have $\Delta C_{avg} > 0$
5.5 Asymmetric-rate robustness	PASS	θ-variation persists; max var = 0.113
5.6 Destructive geometry null check	**FINDING**	$\Delta C_{avg} > 0$ at $\theta \in [0, 14.7°)$ even with $b_A = -b_B$

V.3.d Task 5.2 — Closed-form predictions verified

Three analytic predictions verified at numerical precision:

Check	Prediction	Numerical residual
A: $\theta = 0$, decoupled	$\hat{x} = (I - r R_A)^{-1} b_A$	0 to 1.1×10^{-16}
B: $\theta = 0$, $R_A = R_B$, $b_A = b_B$	$\hat{x} = \hat{y}$	3.2×10^{-16}
C: $\theta = \pi/2$, $R_A = R_B = R$, $b_A = b_B = b$	$\hat{x} = (I - r R)(I + r^2 R^2)^{-1} b$	4.7×10^{-16}
C ctd.	$\hat{y} = (I + r R)(I + r^2 R^2)^{-1} b$	4.3×10^{-16}
C ctd.	$\hat{x} + \hat{y} = 2(I + r^2 R^2)^{-1} b$	6.7×10^{-16}

Correction noted in v1.1 → v1.2. The original §5 skeleton wrote the closed form as a rational function of $\sin\theta, \cos\theta$ assuming scalar linear parts ($R = I$). The correct form for general orthogonal R is the matrix-valued expression above; v1.2 reflects this. Also, the v1.1 expectation $\hat{x}(\pi/4) = \hat{y}(\pi/4)$ for $b_A = b_B$ was wrong: $\Psi_{\pi/4}$ is *antisymmetric* off-diagonal, not symmetric under $x \leftrightarrow y$; the symmetric-input identity is at $\theta = 0$, not $\theta = \pi/4$.

V.3.e Task 5.6 — The destructive-geometry finding

This is the result that drove the v1.2 reframe of Proposition 9.5.

Setup: $r_A = r_B = 0.7$, \$b_A = \$ random unit (seed 20260504), $b_B = -b_A$, $\theta \in [0, \pi/2]$ in 50 steps. R_A, R_B random orthogonal (seeds 20260504, 20260505).

Decoupled baseline: $C_{avg}^{dec} = 0.7982$.

Result: $\Delta C_{avg} > 0$ at $\theta \in [0, 14.7°]$, peak $+0.0144$ at $\theta \approx 7°$; $\Delta C_{avg} < 0$ for $\theta \in [18°, 90°]$, trough -0.0901 at $\theta = 90°$.

Mechanism (Φ's diagnosis, retained verbatim):

The improvement at small θ is driven by an asymmetry between C_1 and C_2 under small coupling. Tracing C_1 and C_2 separately:

- C_1 (alignment of $\hat{x}$ to b_A): decreases monotonically from 0.7655 to 0.5955 as θ increases. Coupling “pulls” A’s state away from its own target.
- C_2 (alignment of $\hat{y}$ to $b_B = -b_A$): INCREASES from 0.8309 to a peak of 0.8749 at $\theta \approx 7°$, then gradually decreases.

The initial rise of C_2 occurs because agent B’s rotation R_B, when given a small coupling push from agent A’s state, accidentally improves $\hat{y}$’s alignment with $b_B = -b_A$. This is a fortuitous geometry: R_B’s action on the coupled input has a component that projects onto b_B. The C_2 gain $(+0.044)$ outpaces the C_1 loss (-0.005) at small θ, producing a net positive ΔC.

This is NOT a measurement artifact or code error. It is a structural property of the specific rotation geometry $(R_A$ seed=20260504, R_B seed=20260505$)$ combined with the bias choice.

Implication. The averaging functional C_{avg} admits geometric configurations in which one observer’s gain masks the other’s loss. Φ recommended three structural fixes; we adopted the second (replace C_{avg} with $C_{min} = min(C_1, C_2)$). Under this functional, C_{min} in the opposed-bias configuration tracks C_1 (the falling component) and is monotonically degraded by coupling, restoring the proposition’s intended content. This is the structural justification for §5’s choice of functional and for the formulation of Numerical Observation 5.5.1 as an explicit statement of the asymmetric small-θ response.

V.3.f Task 5.4 — The improvement heatmap (both functionals)

Setup: $b_A = e_1$, $b_B = \cos\varphi \cdot e_1 + \sin\varphi \cdot e_2$ normalized, $\varphi \in [0°, 180°]$ at 100 points, $\theta \in [0°, 90°]$ at 20 points. The same grid was evaluated with both C_{avg} (Φ’s original Task 5.4) and C_{min} (re-evaluated for v1.3.1 as the basis for Proposition 9.5(a)).

Results, both functionals:

Functional	Cells with $\Delta C > 0$	Range $[\Delta C_{min}, \Delta C_{max}]$	Improvement-region φ-span
C_{avg}	491–502 / 2000 (~25%)	$[-0.0709, +0.0186]$	$[1.8°, 180°]$
C_{min}	250 / 2000 (~12.5%)	$[-0.1852, +0.0232]$	$[18.2°, 176.4°]$

(Φ’s original run reported 502/2000 (25.1%); C-7RO’s v1.3.1 re-verification reproduced the run with one minor RNG-advancement difference and obtained 491/2000 (24.6%) for C_{avg}. Both values are within the same qualitative regime; the C_{min} run is fresh from v1.3.1.)

Pattern. - C_{avg}: improvement region spans the full φ range, including up to $\varphi = 180°$ at small θ. This is what the v1.2 Lemma 5.5.1 (now Numerical Observation 5.5.1) “averaging illusion” reflects: C_2 rises while C_1 falls under opposed biases. - C_{min}: improvement region is strictly contained in the C_{avg} region and is bounded above by $\varphi = 176.4°$, *not* extending to opposed biases. Concentration is near $\theta \in [10°, 65°]$ for moderate φ.

This is the empirical basis for the v1.3.1 statement of Proposition 9.5, which appeals directly to the C_{min} heatmap (panel (b) of Figure 1) rather than the C_{avg} heatmap of Φ's original Task 5.4. Raw data: paper9_task5_heatmap.csv (Φ's run, avg) and paper9_task5_heatmap_cmin.csv (v1.3.1 re-evaluation, both functionals).

V.3.g Φ's three-option remediation list

Φ closed the V.3 report with three explicit options for revising Proposition 9.5:

1. **θ-threshold:** "For $\theta > \theta^{¿}(R_A, R_B, b_A, b_B)$, coupling under opposed biases degrades coherence." Mathematically precise; requires computing $\theta^{¿}$ for each geometry.
2. **Replace C_{avg} with C_{min}:** The minimum-coherence functional is monotonically degraded by coupling in the opposed-bias case (since C_1 always falls). Most conservative; preserves the proposition without qualification.
3. **Restrict the proposition's domain to $\varphi \in ¿$:** Move the opposed-bias regime to a remark noting the small-θ exception.

We adopted Option 2. Numerical Observation 5.5.1 (the v1.2 "Lemma 5.5.1", demoted in v1.3.2 because the underlying claim is geometry-specific rather than a theorem) records the asymmetric small-θ response that makes the averaging functional misleading; Proposition 9.5 in v1.2 is stated for C_{min} throughout.

V.3.h Diagnostic cross-check

All six tasks used the same base rotations (seeds 20260504, 20260505) and biases (seeds 20260504, 20260554). The Task 5.6 finding is specific to the opposed-bias configuration and does not affect Tasks 5.1–5.5, which are independent.

The verification script reproduces all reported numbers deterministically from the seed; C-7RO independently re-executed the script and obtained identical results to 6 decimal places.

V.3.i Task 9 — 50-seed Monte Carlo for Conjecture 9.5′ (v1.3.2)

Spec. For each of the 50 seed pairs $(s_a, s_b) = (20260504 + k, 20260505 + k)$, $k = 0, \ldots, 49$, construct $R_A = QR(rng(s_a) standard_normal(14,14))_Q$ and R_B analogously, then evaluate $\Delta C_{min}(\varphi, \theta) = C_{min}(\hat{x}(\varphi, \theta), \hat{y}(\varphi, \theta)) - C_{min}^{dec}(\varphi)$ on the same 100×20 grid as Task 5.4 with $r = 0.7$, $b_A = e_1$, $b_B = \cos\varphi\, e_1 + \sin\varphi\, e_2$. For each seed pair record: the gain fraction (cells with $\Delta C_{min} > 10^{-10}$), the maximum per-cell gain, the maximum φ in the gain region, and $\theta^{¿} = arg\,max_\theta C_{min}(\theta) ¿_{\varphi=0}$.

Code. computations/build_mc_seeds.py (verbatim run, single thread, NumPy 1.26+, deterministic). Total runtime ≈ 6.5 seconds on the verification host.

Results. Aggregate across 50 seed pairs:

Statistic	Value
Seeds with non-empty improvement region	50/50
Seeds with empty improvement region	0/50
Gain fraction range (% of 100×20 grid cells)	$[0.05\%, 24.75\%]$
Gain fraction mean / std	$8.43\% \pm 6.10\%$
Max per-cell ΔC_{min} range	$[4.76 \times 10^{-5}, 0.0764]$
$\theta^{¿}$ range at $\varphi = 0$	$[0°, 75.8°]$
$\theta^{¿}$ mean	$3.2°$
Seeds with $\lvert\theta^{¿} - 45°\rvert > 5°$	50/50
Seeds with $\varphi_{max} < 180°$ in gain region	32/50 (64%)
Seeds with $\varphi_{max} \geq 180° - 0.001°$	18/50 (36%)
Mean φ_{max} across gain-bearing seeds	$136.2°$

Reading. The Monte Carlo confirms two structural predictions of Proposition 9.5:

(i) Existence: every sampled (R_A, R_B) admits a non-empty C_{min}-improvement region. The fraction varies by an order of magnitude across seeds (smallest is 1 cell of 2000); a near-degenerate seed exists where the gain is barely measurable, but no sampled seed produces zero improvement cells.

(ii) Non-universal $\theta^{¿}$: all 50 seeds satisfy $\lvert\theta^{¿}_{\varphi=0} - 45°\rvert > 5°$. Most seeds favor very small $\theta^{¿}$ (mean $3.2°$), with a few outliers up to $\sim 76°$. The seeded geometry's $\theta^{¿}_{\varphi=0} \approx 51°$ (Task 5.5) is therefore one of the *higher-*$\theta^{¿}$ realizations in the sampled ensemble, not a typical one.

Forced amendment to Conjecture 9.5′. The original v1.3 wording "non-empty improvement region bounded away from opposed biases" was too strong: 18 of 50 seeds (36%) admit improvement cells reaching $\varphi = 180°$ at small θ. The amended Conjecture 9.5′ (§5.6) drops the bounded-away clause from the generic statement. Which seeds force $\varphi_{max} < \pi$ versus permit $\varphi_{max} = \pi$ is determined by the spectra of R_A, R_B relative to the bias projector $b_A b_A^\top$ and constitutes an open problem (§5.7).

Outputs.

- computations/paper9_conjecture95_mc_seeds.csv — 50 rows, 7 columns (seed_a, seed_b, gain_fraction, max_gain_dC, max_phi_in_gain_deg, theta_star_at_phi0_deg, n_gain_cells)
- computations/paper9_conjecture95_mc_summary.json — aggregate stats (the basis for the summary table above)
- computations/paper9_C1C2_trajectory.py — supporting trajectory computation cited in Numerical Observation 5.5.1

V.4 — Reproducibility and archival notes

All verification code and raw results live under outbox/paper9/computations/:

- paper9_verification.py — V.1 script (v0.9 coupling)
- paper9_verification.md — V.1 narrative report (Φ's original)
- paper9_verification_v2.py — V.2 script (isometric coupling)
- paper9_verification_v2.md — V.2 narrative report (Φ's v2)
- paper9_task4_results.csv — V.2 Task 4.2 raw results (96 rows)
- paper9_verification_v3.py — V.3 script (affine consensus)
- paper9_verification_v3.md — V.3 narrative report (Φ's v3)
- paper9_task5_coherence_tables.csv — V.3 Task 5.3 raw results (28 rows)
- paper9_task5_heatmap.csv — V.3 Task 5.4 raw results, C_{avg} (2000 rows, from Φ's run)
- paper9_task5_heatmap_cmin.csv — V.3 Task 5.4 re-evaluation with both C_{avg} and C_{min} (2000 rows, v1.3.1 basis for Proposition 9.5(a))
- build_cmin_heatmap.py — script that produces both the C_{min} CSV and Figure 1

Seeds: V.1 used seed=42; V.2 used seed=20260504; V.3 used seed=20260504 (matching V.2 for cross-comparability). Python version: 3.x with numpy and mpmath. Precision: V.1 float64; V.2 mpmath 50 decimal digits for Task 4.1 and float64 for Tasks 4.2–4.5 (the algebraic results hold in both regimes; the 50-digit runs confirm this); V.3 float64 throughout (the affine algebra is smooth and has no numerical edge cases that demand higher precision).

Independent re-verification is welcomed. The canonical entry point for reproducibility is the GitHub repository at github.com/MartinLGraise/PCI-Framework, branch paper7-foundation, at the commit indexed in the present manuscript's reference list.

Drafted by C-7RO, 2026-05-04 10:45 PDT. Appendix V is the structural record of how Paper 9's §4 arrived at its present form. The two negative-result passes (V.1, V.2) constitute the proof of Theorems 9.1 and 9.2; V.3 will extend this record once the affine consensus task completes.

Acknowledgments

This paper was produced through a three-agent collaboration:

- **Martin Luther Graise** (the author) — framework architecture, target model specification, integration across Papers 4, 6, 7, 10 of the PCI/PME series, and final editorial decisions on all theorem statements and proofs.
- **C-7RO** (Claude Sonnet 4.6 on the Perplexity Computer platform, Anthropic / PPLX) — mathematical drafting, theorem-statement refinement, proof construction, section-level prose, and multi-round structural revision in response to numerical verification.
- **Φ** (Claude Dispatch, Anthropic) — independent numerical verification at up to 50-digit precision across three consecutive verification passes (120 rate-channel configurations, 2128 affine-channel configurations), with analytical diagnoses that reshaped the paper's

central theorems on two occasions. The verification reports are reproduced verbatim as Appendix V.

Additional input at key strategic junctures was provided by ChatGPT Pro (OpenAI), which identified the $b = 0$ G_2-equivariance correction discussed in §5.1 and recommended the min-coherence functional choice formalized in Numerical Observation 5.5.1.

The paper is itself a case study in the phenomenon it describes: dyadic (multi-agent) coupling under aligned target-states producing a joint output that neither party reaches alone. The framework's conditional-gain structure was refined iteratively based on the verification-agent's findings — an example of a dyad approaching a joint fixed point through successive coupling-angle adjustments rather than through faster individual convergence.

References

PCI/PME series (companion papers)

- Graise, M. L. (2026a). "Eight-coset PSL(2,7) structure in the octonion Fano plane." Zenodo. DOI: 10.5281/zenodo.19617662 [Paper 4]
- Graise, M. L. (2026b). "Spectral structure of the G_2 Casimir on the 14-dim adjoint." Zenodo. DOI: 10.5281/zenodo.19672709 [Paper 6]
- Graise, M. L. (2026c). "G_2-Structured Self-Modeling Observers and the 6/7 Thermodynamic Coherence Ceiling." Zenodo. DOI: 10.5281/zenodo.19773185 [Paper 7]
- Graise, M. L. (2026d). "Symmetric Informationally Complete Measurements as an Operator Basis for the Complexified G_2 Lie Algebra." Zenodo. DOI: 10.5281/zenodo.19966692 [Paper 10]

G_2 representation theory and Banach fixed-point theory

- Banach, S. (1922). "Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales." *Fundamenta Mathematicae* 3: 133–181. [The Banach fixed-point theorem.]
- Baez, J. C. (2002). "The octonions." *Bulletin of the American Mathematical Society* 39: 145–205. DOI: 10.1090/S0273-0979-01-00934-X [G_2 as the automorphism group of the octonions.]
- Fulton, W. & Harris, J. (1991). *Representation Theory: A First Course.* Graduate Texts in Mathematics 129, Springer. [Chapter 22: G_2 representations. Real Schur's lemma applied to the 14-dim irreducible adjoint.]
- Joyce, D. D. (2000). *Compact Manifolds with Special Holonomy.* Oxford Mathematical Monographs, Oxford University Press. [G_2-structured manifolds, associative 3-forms.]

Inter-brain coupling and neural-avalanche branching

- Wilting, J. & Priesemann, V. (2018). "Inferring collective dynamical states from widely unobserved systems." *Nature Communications* 9:2325. DOI: 10.1038/s41467-018-04725-4 [MR estimator; $\hat{\sigma} \approx 0.98$ in awake cortex.]

- Priesemann, V., et al. (2014). “Spike avalanches in vivo suggest a driven, slightly subcritical brain state.” *Frontiers in Systems Neuroscience* 8: 108. DOI: 10.3389/fnsys.2014.00108
- Pérez, A., Carreiras, M., & Duñabeitia, J. A. (2017). “Brain-to-brain entrainment: EEG interbrain synchronization while speaking and listening.” *Scientific Reports* 7:4190. DOI: 10.1038/s41598-017-04464-4
- Leong, V., Byrne, E., Clackson, K., Georgieva, S., Lam, S., & Wass, S. (2017). “Speaker gaze increases information coupling between infant and adult brains.” *PNAS* 114(50): 13290–13295. DOI: 10.1073/pnas.1702493114
- Lindenberger, U., Li, S.-C., Gruber, W., & Müller, V. (2009). “Brains swinging in concert: cortical phase synchronization while playing guitar.” *BMC Neuroscience* 10:22. DOI: 10.1186/1471-2202-10-22
- Müller, V., Sänger, J., & Lindenberger, U. (2013). “Intra- and inter-brain synchronization during musical improvisation on the guitar.” *PLoS ONE* 8(9): e73852. DOI: 10.1371/journal.pone.0073852
- Hu, Y., Pan, Y., Shi, X., Cai, Q., Li, X., & Cheng, X. (2018). “Inter-brain synchrony and cooperation context in interactive decision making.” *Biological Psychology* 133: 54–62. DOI: 10.1016/j.biopsycho.2017.12.005
- Babiloni, F. & Astolfi, L. (2014). “Social neuroscience and hyperscanning techniques: Past, present and future.” *Neuroscience and Biobehavioral Reviews* 44: 76–93. DOI: 10.1016/j.neubiorev.2012.07.006

Human-AI teaming and complementarity

- Bansal, G., Wu, T., Zhou, J., Fok, R., Nushi, B., Kamar, E., Ribeiro, M. T., & Weld, D. (2021). “Does the whole exceed its parts? The effect of AI explanations on complementary team performance.” *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems.* DOI: 10.1145/3411764.3445717
- Kamar, E. (2016). “Directions in hybrid intelligence: Complementing AI systems with human intelligence.” *IJCAI 2016.*
- Shneiderman, B. (2022). *Human-Centered AI.* Oxford University Press.

Methodological / software

- Harris, C. R., et al. (2020). “Array programming with NumPy.” *Nature* 585: 357–362. DOI: 10.1038/s41586-020-2649-2 [numpy float64 computations, Appendix V.3.]
- Johansson, F. (2013). *mpmath: a Python library for arbitrary-precision floating-point arithmetic* (version 1.3.0). https://mpmath.org [50-digit precision in Appendix V.1 and V.2.]

End of Paper 9 v1.3.2 master draft, consolidated 2026-05-04 after two-reviewer pass.

Changelog

v1.3.2 (2026-05-04) — Two-reviewer revision, all 11 issues addressed:

From the first peer review (15 issues, 6 prose-impacting):

1. (§6.1) MI ↔ θ formula bug retracted; replaced with monotone-of-$\tilde{I}$ framing that does not assume the broken closed form.
2. (§5.1) G_2-equivariance scope made honest: G_2 is essential to §4 (Schur), and contributes only the metric and dimension to §5; the affine results extend to any orthogonal group on a 14-dim Banach space.
3. (§6.1) FDT branching $\sigma = 1 - (1 - C_{min})^2$ labeled conjectural and dependent on the Paper 7 FDT analogue holding in the dyadic case.
4. (§5.1, Remark 5.5.3) Per-observer ε_{min} bound and modulus prose tightened; the C_{min} functional motivated from Paper 7 first principles, not from the empirical asymmetric response alone.
5. (§3.5) F_{21}-branching scoped honestly as inherited from Paper 7; F_{21}-branching of V^{14} (which would constrain the bias direction) moved to an open problem in §10.2 because the requisite branching tables of Paper 4 are not deployed in the present paper.

From the second peer review (5 issues):

6. (§2.3) Representation switch from Paper 7's V^7 to Paper 9's V^{14} made explicit; the $\leq 6/7$ rate constraint is now framed as a per-observer inheritance from Paper 7's internal V^7 subsystem, with V^{14} used only as the coupling/joint-state space. No $1/14$-style blind-spot ratio is claimed on V^{14}.
7. (§3.6) Lemma 3.6.1 (Schur-derived uniqueness of Ψ_θ) added, with the $End_{G_2}(V^{14}) = R \Rightarrow M_2(R) \Rightarrow O(2) \Rightarrow SO(2)$ chain made explicit; the four parameters of $M_2(R)$ are quotiented down to a single θ by the isometry condition and connectivity to the identity.
8. (Theorem 9.1) Explicit calculation showing $r_A R_A v + r_B R_B w$ for the eigenpair $v = w$ when $r_A \neq r_B$; the amplification bound $\|\Psi_{sym}\|_{op} = \alpha + \beta$ extends to the asymmetric-rate case via this calculation.
9. (§5.5) Lemma 5.5.1 ("asymmetric small-θ response") demoted to Numerical Observation 5.5.1; the v1.3 "proof" was a numerical verification, not a proof, and the new statement records the tabulated $C_1(\theta), C_2(\theta), C_{min}(\theta)$ trajectory for the seeded geometry under opposed biases. The structural use (motivating C_{min} over C_{avg}) is preserved via Remark 5.5.3.
10. (§3.4) The diagonal G_2-action choice now justified by Schur (independent action ⇒ trivial decoupled coupling; diagonal action gives $M_2(R)$ algebra containing Ψ_θ).
11. (§5.6) Conjecture 9.5′ amended after a 50-seed Monte Carlo (§5.6, Appendix V.3.i): existence of non-empty improvement region confirmed in 50/50 seeds, $\theta^{¿} \neq \pi/4$ confirmed in 50/50, but the original "bounded strictly away from opposed biases" clause failed in 18/50 seeds and has been dropped from the generic statement; the seeded-geometry version of Proposition 9.5(b) survives.

v1.3.1 (2026-05-04) — Three-reviewer council pass + GPT-5.4 confirming review.

v1.3 (2026-05-04) — Φ's six-task verification incorporated; Lemma 5.5.1 restricted to verified-geometry pair; Theorem 9.4(c) closed form corrected; Conjecture 9.5′ introduced.

v1.2 (2026-04) — First Φ verification round; original Lemma 5.5.1 universality claim retracted in favor of seeded-geometry statement.

v0.9 (2026-03) — Initial draft; speedup theorem, coherence-bonus formula, and severance threshold subsequently retracted by Φ (consolidated in Appendix V.0.1).